

Lecture 5.2: WiFi Sensing

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Contents

- Channel State Information
- Multipath Effect
 - Reflection Model
 - Scattering Model
- Geometrical Approaches
- Statistical Approaches
 - Speed Estimation
 - Motion Detection
 - Breathing Rate Estimation
- More Applications

WiFi Sensing

- “The next big wireless movement” – By Ray Liu, IEEE President, Founder & CEO of Origin AI



Multipath Everywhere!

- 🏷️ Motion Detection
- 🏷️ Sleep Monitoring
- 🏷️ Fall Detection
- 🏷️ Gait Recognition
- 🏷️ Gesture Control
- 🏷️ Wellbeing Monitoring
- 🏷️ Activity Monitoring
- 🏷️ Location Tracking
- 🏷️ Many More...

WiFi Sensing interprets the multipath disruptions for sensing.

No Cameras, No Wearables, No Sensors

**Wireless
Contactless
Sensorless**



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Review: What is CSI?

A data perspective w/ zero SP background & zero memory about previous lectures

- $\mathbf{H}(\mathbf{t}) = [H(t, f_1), H(t, f_2), \dots, H(t, f_N)]$
 - Complex number: $H(t, f_i) = a_i + jb_i$
 - Amplitude: $|H(t, f_i)| = \sqrt{a_i^2 + b_i^2} \quad \leftarrow \text{abs}()$
 - Phase: $\phi_i = \tan^{-1} \frac{b_i}{a_i} \quad \leftarrow \text{phase}()$

- Time series of $\mathbf{H}(\mathbf{t})$

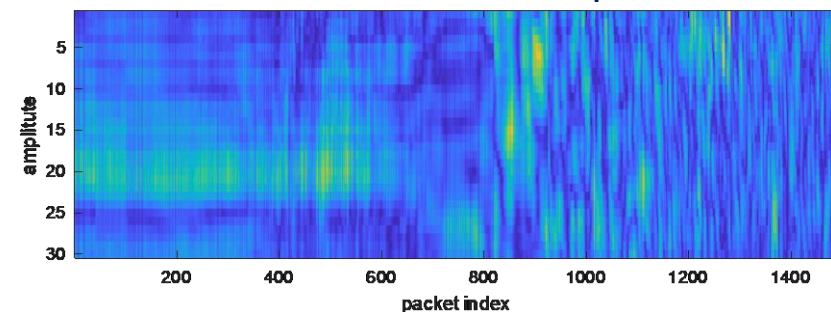
Time Series of CSI

$$\begin{bmatrix} H(1, f_1) & \cdots & H(M, f_1) \\ \vdots & \ddots & \vdots \\ H(1, f_N) & \cdots & H(M, f_N) \end{bmatrix}$$

Time Series of CSI Amplitude

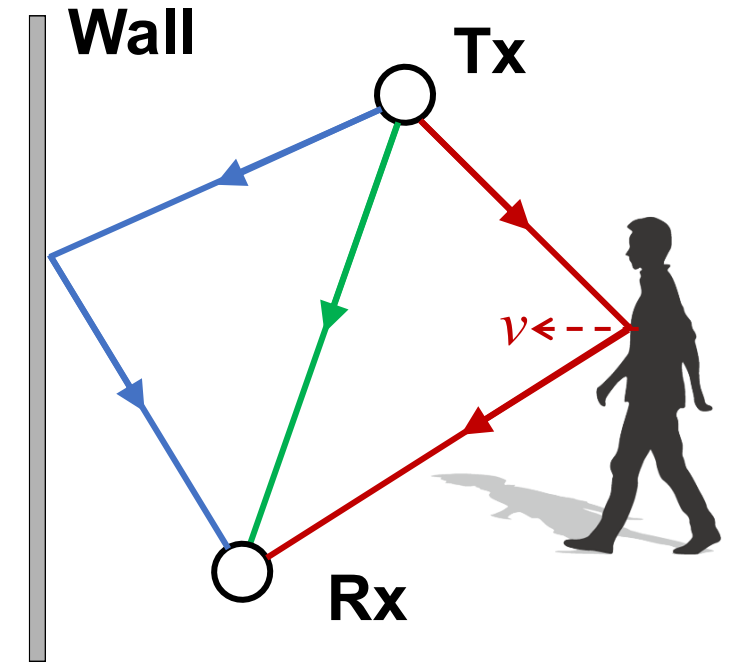
$$\begin{bmatrix} |H(1, f_1)| & \cdots & |H(M, f_1)| \\ \vdots & \ddots & \vdots \\ |H(1, f_N)| & \cdots & |H(M, f_N)| \end{bmatrix}$$

Time Series of CSI Amplitude

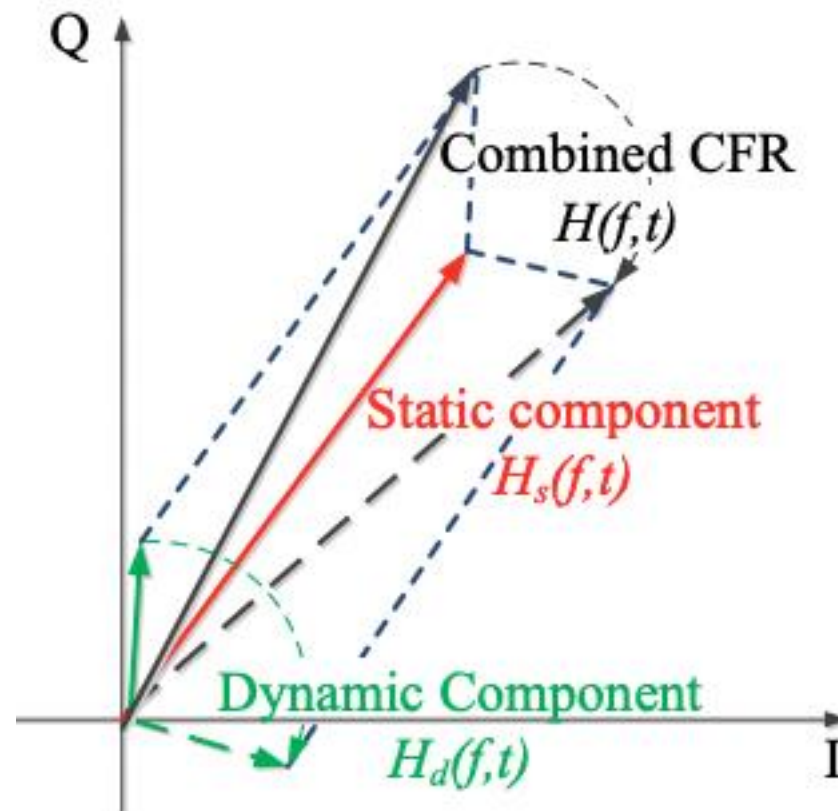
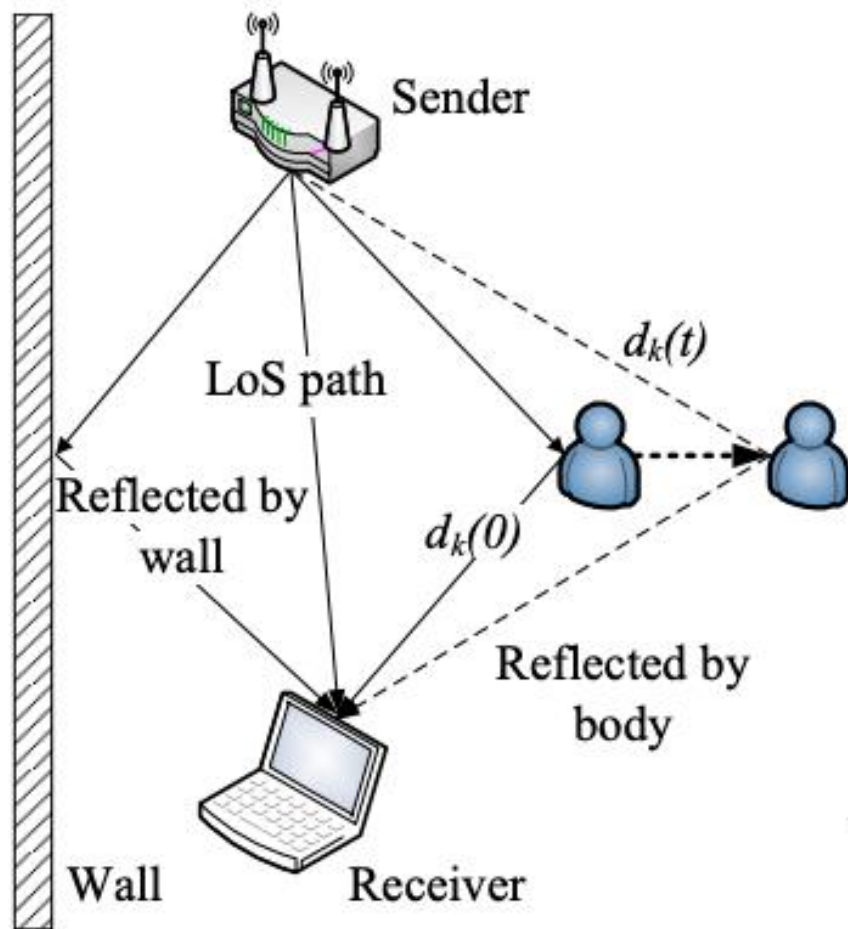


Geometric Parameters

- Time of Flight (ToF)
 - Range
 - Time Difference of Arrival (TDoA)
 - Angle of Arrival (AoA)
 - Doppler Frequency Shift (DFS)
 - Velocity
-
- Resolve the above parameters of (major) multipath signals



CSI Dynamics



CSI Dynamics

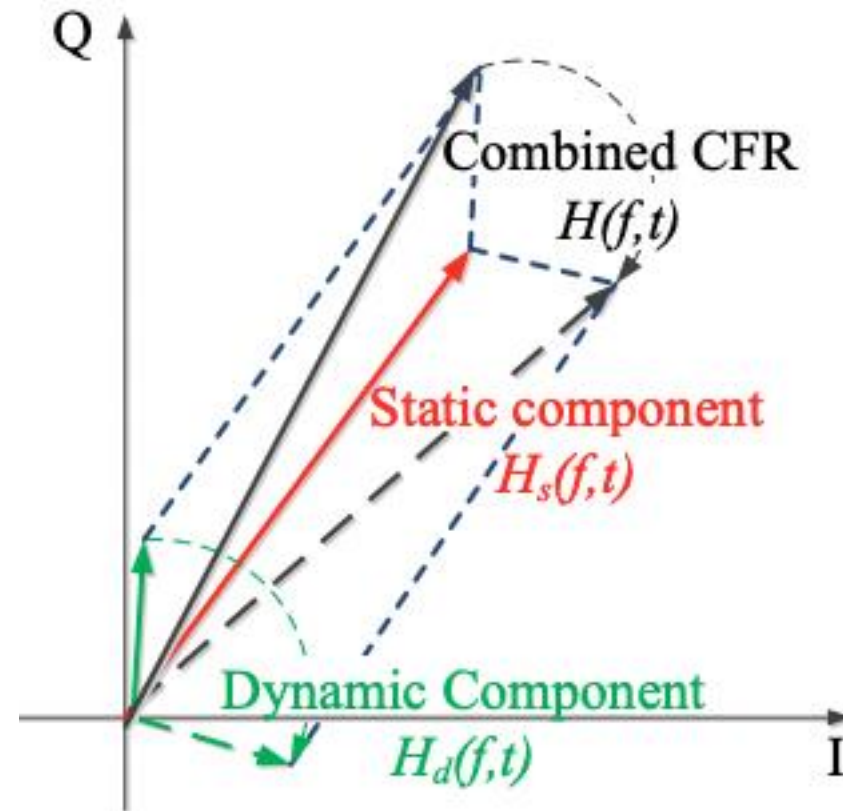
$$H(t, f) = \sum_{l \in \Omega} a_l(t) \exp(-j2\pi f \tau_l(t))$$

$$H(f, t) = \underline{H_S(f, t)} + \underline{H_D(f, t)}$$

Static components

Dynamic components

$H_D(f, t)$ has approximately zero mean
→ Therefore, $H(f, t)$ has approximately zero mean by removing the static part $H_S(f, t)$.



CSI Dynamics to DFS

- Doppler Frequency Shifts

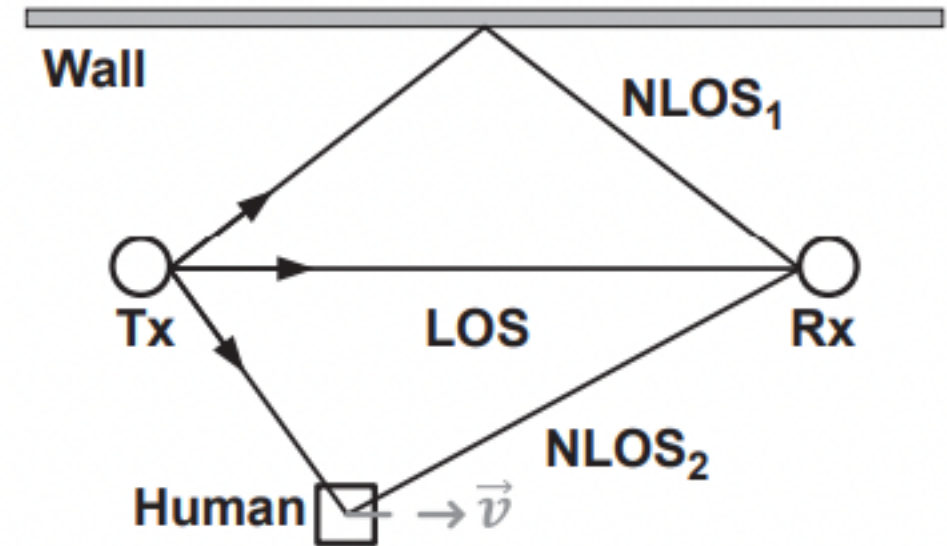
- Caused by (reflection) path length changes

- $f_D(t) = -\frac{1}{\lambda} \frac{d}{dt} d(t) = -f \frac{d}{dt} \tau(t)$

Path Length Change
Rate (PLCR)

Ideal CSI:
$$H(f, t) = H_S(f, t) + \sum_{l \in \Omega_D} \alpha_l(t) e^{j2\pi f \tau_{\infty} + f D_l(u) du}$$

Measured CSI:
$$\hat{H}(f, t) = H(f, t) e^{-j2\pi(\Delta_t f + \Delta_f t)}$$



CSI to DFS

- CSI at frequency (subcarrier) f , at time t with K propagation paths

$$H(f, t) = \sum_{k=1}^K \alpha_k(t) e^{j2\pi f \tau_k(t)}$$

amplitude of path k at time t (points to $\alpha_k(t)$)
 propagation delay of path k at time t (points to $\tau_k(t)$)
 phase of path k (points to the exponent $j2\pi f \tau_k(t)$)

- In case of motions

$$H(f, t) = H_{\text{static}}(f) + \sum_{k \in K_{\text{dynamic}}} \alpha_k(t) e^{j2\pi \int_{-\infty}^t f_{D_k}(u) du}$$

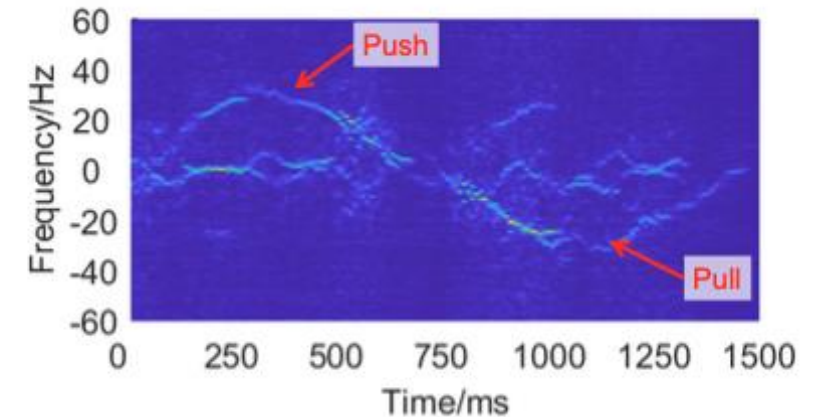
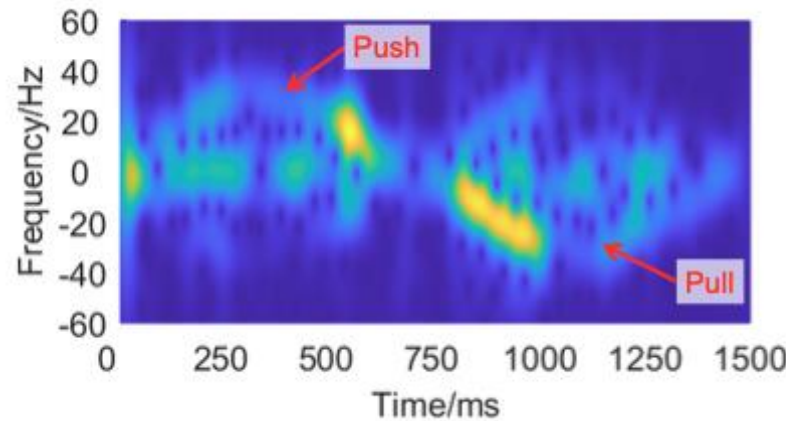
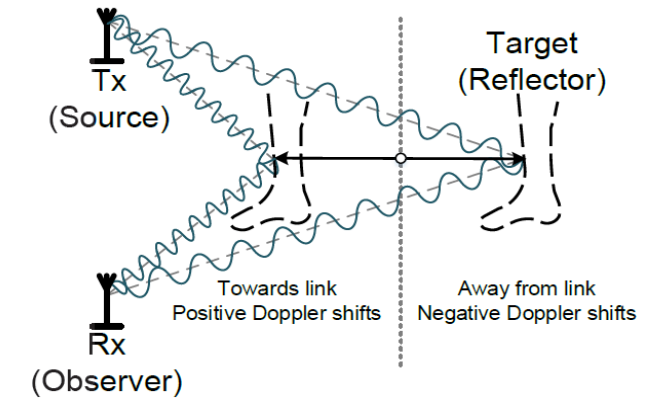
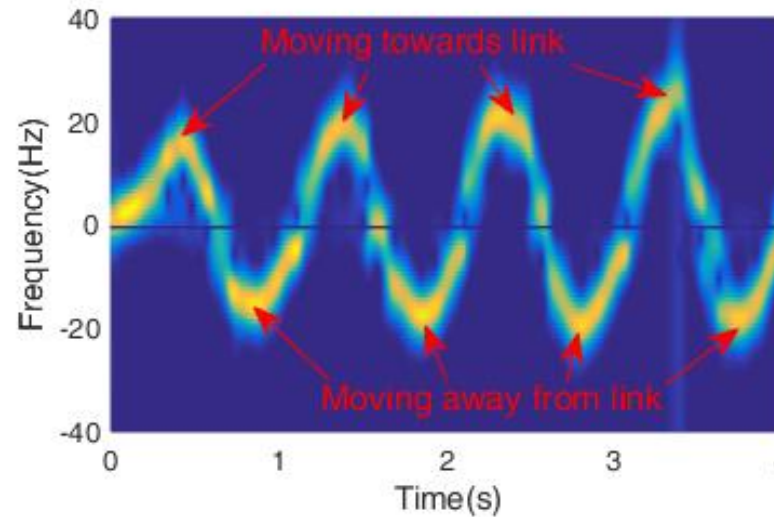
doppler frequency shift of (dynamic) path k (points to $f_{D_k}(u)$)
 Integral of frequency over time $\propto$ phase (points to the integral term)
 CSI of static paths (points to $H_{\text{static}}(f)$) CSI of dynamic paths (points to the sum term)

CSI to DFS: Example

CSI of **dynamic** paths
(spectrogram)

Spectrogram generation by STFT

- Short-time Fourier transform

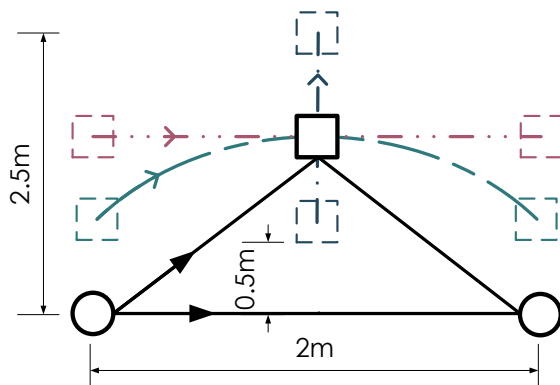


Two practical issues of DFS

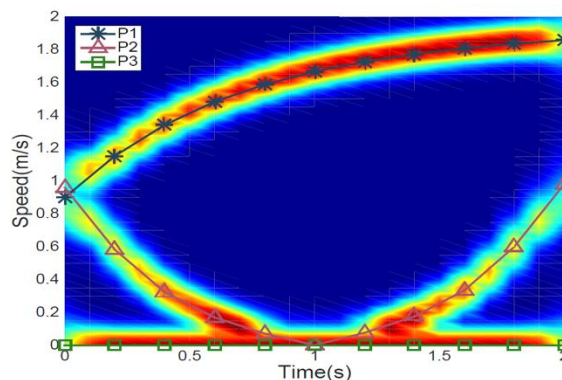
- Issue I: Phase Errors

Ideal CSI:
$$H(f, t) = H_S(f, t) + \sum_{l \in \Omega_D} \alpha_l(t) e^{j2\pi \int_{-\infty}^t f D_l(u) du}$$

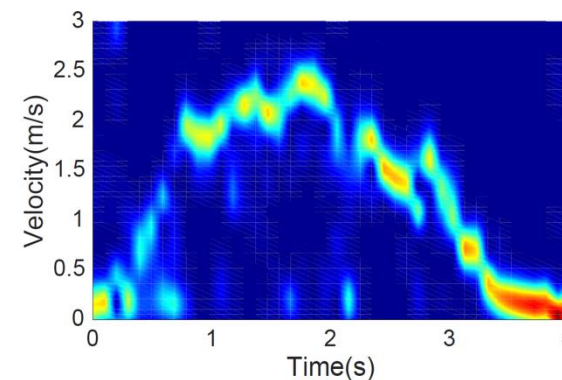
Measured CSI:
$$\hat{H}(f, t) = H(f, t) e^{-j2\pi(\Delta_t f + \Delta_f t)}$$



Walking Scenarios



Ideal Spectrograms



Real Spectrograms

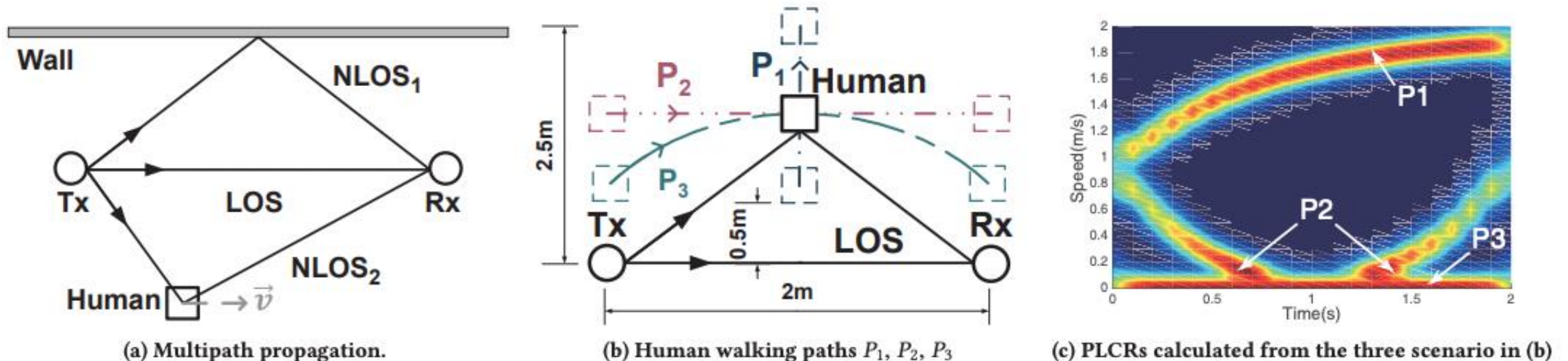
Two practical issues of DFS

- Issue I: Phase Errors
- Solution I (Discard phase): Use CSI power only
 - $|\hat{H}(f, t)|^2 = |H(f, t)|^2$
 - Eliminates the impact of phase errors
 - But also loses the sign of DFS
- Solution II (Phase cleaning): Linear phase fitting
 - Ideally, the phase offsets are linear across subcarriers
- Solution III (Phase cleaning): Antenna difference
 - Different antennas share the same CFO, CTO (but not initial offset)
 - **Conjugate multiplication** of CSI on multiple antennas
 - Needs at least two antennas; initial offset needs manual calibration

Two practical issues of DFS

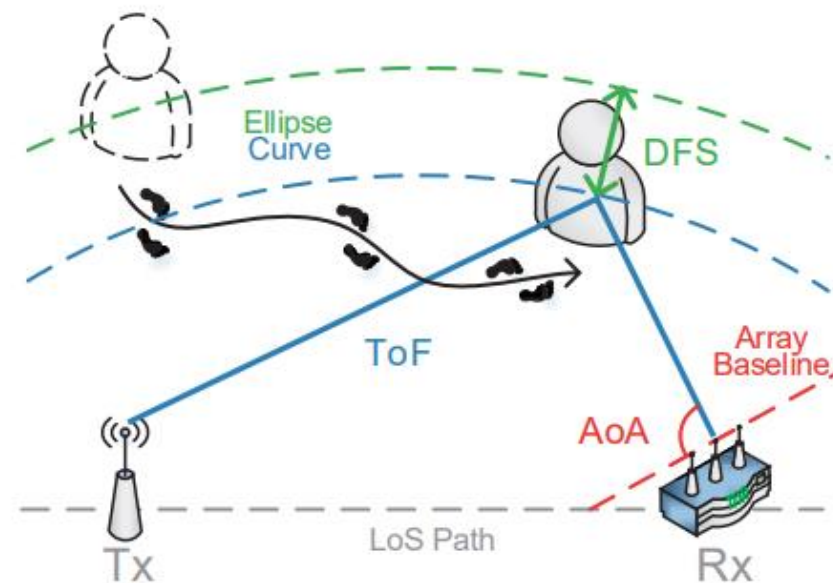
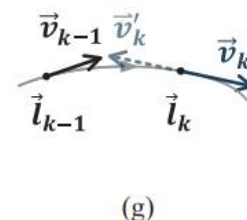
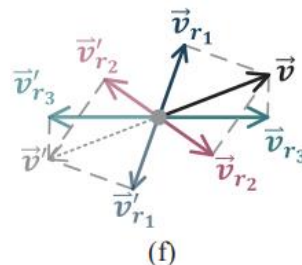
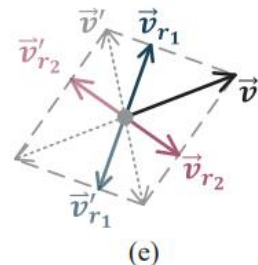
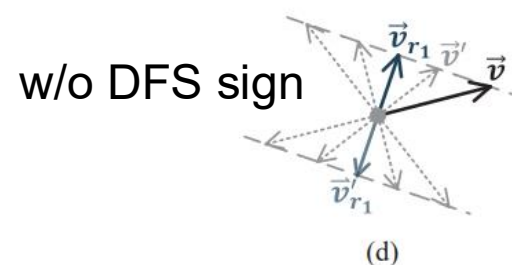
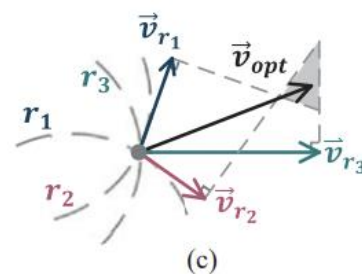
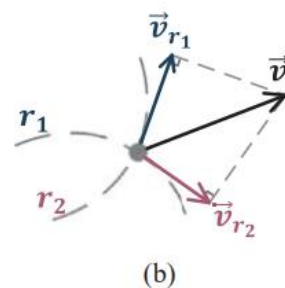
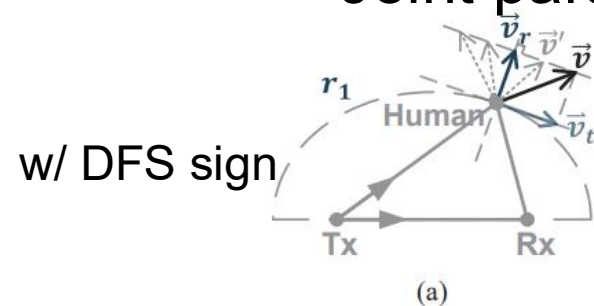
- Issue II: Partial Speed

- DFS from CSI does not reveal complete speed
- Even with accurate phase and thus accurate DFS, the speed depends on the location and moving direction



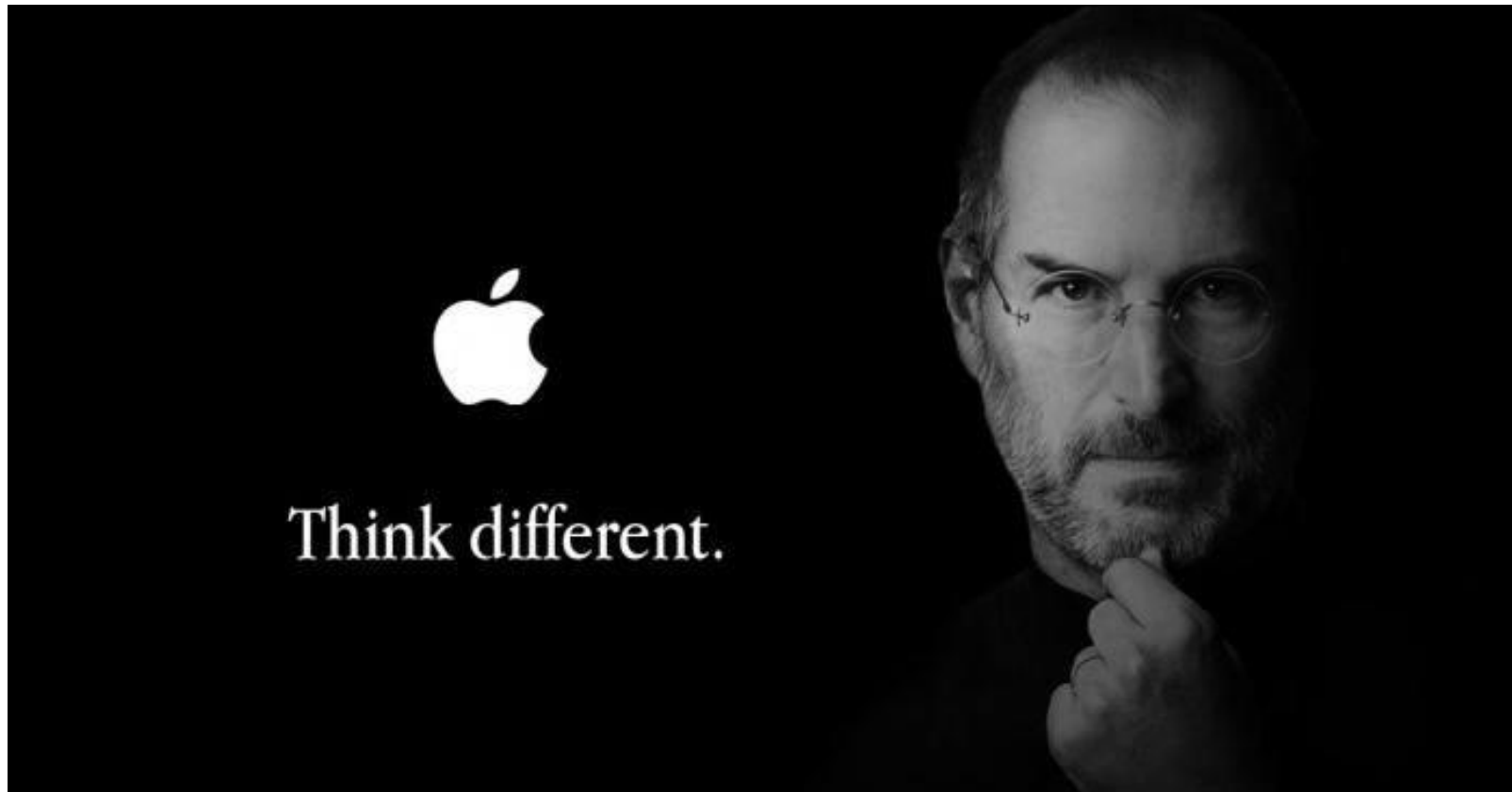
Two practical issues of DFS

- Issue II: Partial Speed
- Solution
 - Fuse information from multiple links
 - Joint parameter estimation (AoA, ToF, DFS, etc)



We need a better model

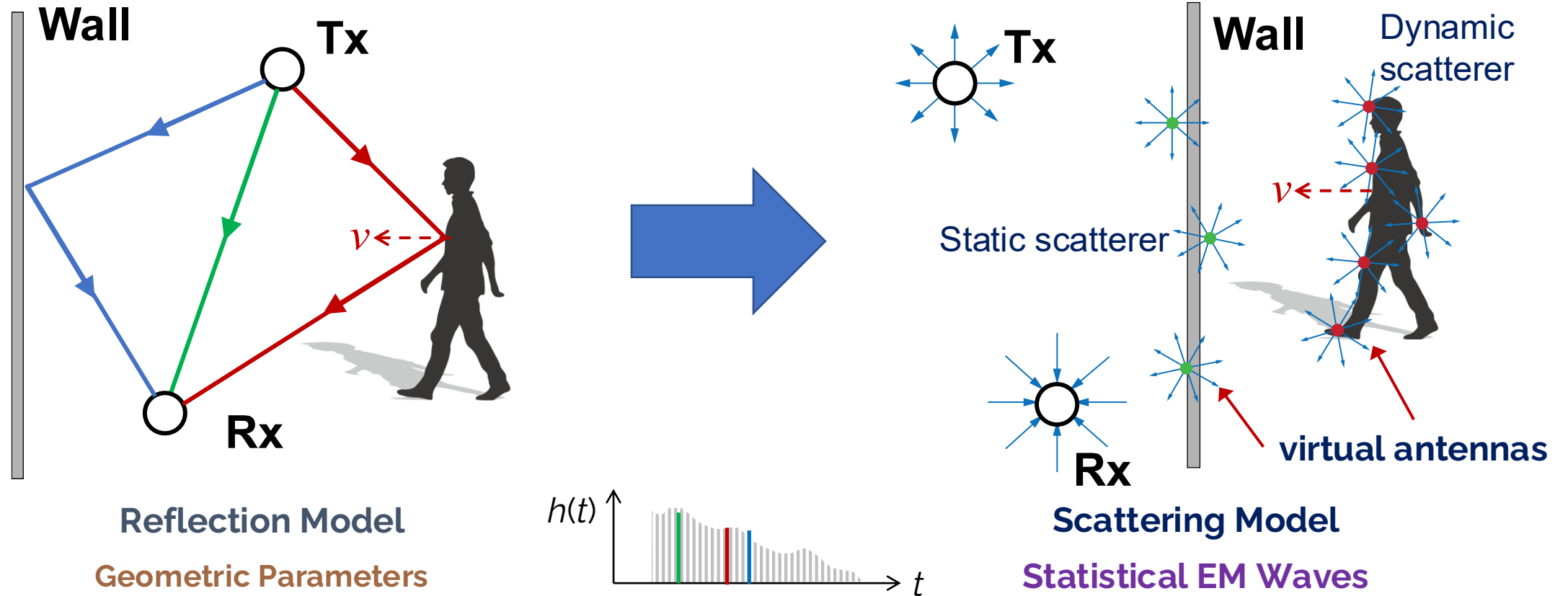
- To overcome these fundamental and practical problems.



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Scattering Model



Multipaths → Scatterers → Virtual Antennas!

Statistical EM Approach

- Decomposition of Received Electric Field

$$\vec{E}_{RX}(t, f) = \underbrace{\sum_{i \in \Omega_s(t)} \vec{E}_i(t, f)}_{\text{Static scatterers}} + \underbrace{\sum_{j \in \Omega_d(t)} \vec{E}_j(t, f)}_{\text{dynamic scatterers}}$$

Static scatterers dynamic scatterers

- Autocorrelation Function (ACF) of $\vec{E}_i(t, f)$:

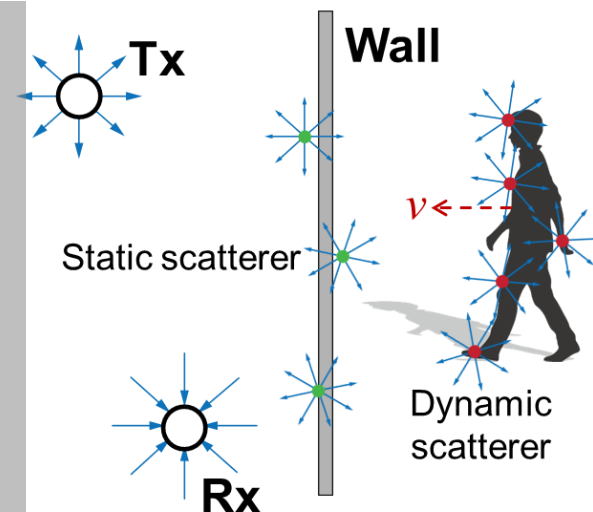
$$\begin{aligned} \rho_{E_{ix}}(\tau, f) &= \rho_{E_{iy}}(\tau, f) \\ &= \frac{3}{2} \left[\frac{\sin(kv_i\tau)}{kv_i\tau} - \frac{1}{(kv_i\tau)^2} \left(\frac{\sin(kv_i\tau)}{kv_i\tau} - \cos(kv_i\tau) \right) \right], \\ \rho_{E_{iz}}(\tau, f) &= \frac{3}{(kv_i\tau)^2} \left[\frac{\sin(kv_i\tau)}{kv_i\tau} - \cos(kv_i\tau) \right]. \end{aligned}$$



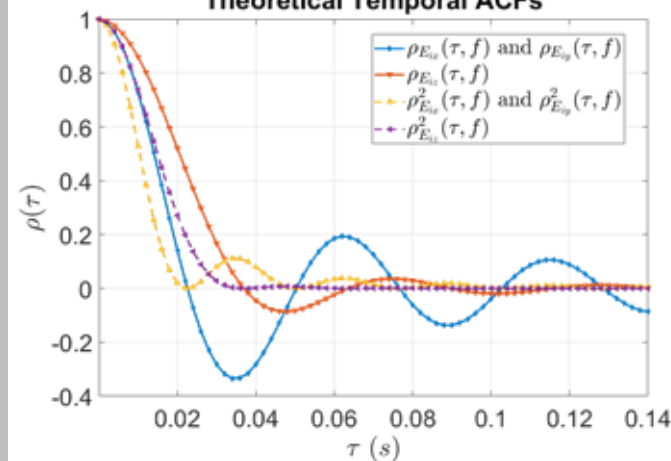
$$\rho_{E_{iu}}(\tau, f) = F_u(kv_i\tau), \forall u \in \{x, y, z\}$$

$k = 1/\lambda$: wavenumber; v_i : speed of scatterer i

The ACF is a function of speed!



Set $v_i = 1 \text{ m/s}$, $f = 5.805 \text{ GHz}$
Theoretical Temporal ACFs



Statistical EM Approach

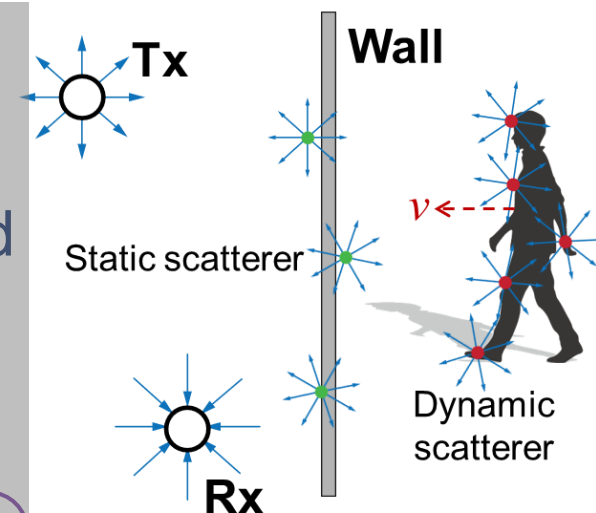
- From $\vec{E}_i(t, f)$ to CSI:
- In practice, $\vec{E}_{Rx}(t, f)$ is a vector and cannot be measured by WiFi directly.
- Instead, the power of $\vec{E}_{Rx}(t, f)$ is measured:

Power of CSI: $G(t, f) \triangleq |H(t, f)|^2 = \|\vec{E}_{Rx}(t, f)\|^2 + \varepsilon(t, f)$

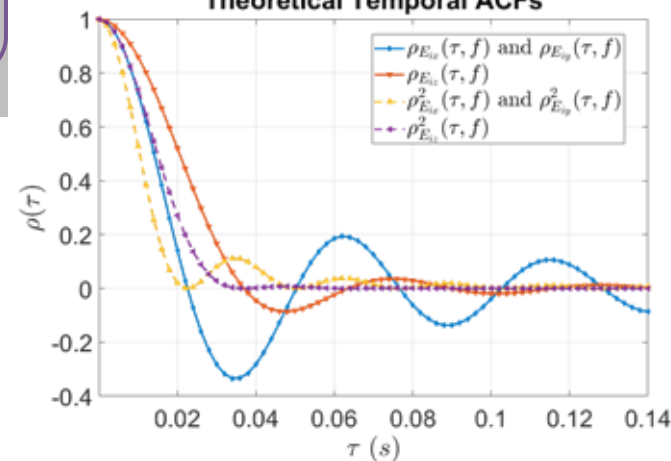
ACF of $G(t, f)$: $\rho_G(\tau, f) \approx C(f) \left(E_d^2(f) \rho_{E_{ix}}^2(\tau, f) + E_s^2(f) \rho_{E_{ix}}(\tau, f) + \delta(\tau) \sigma^2(f) \right)$

The model allows:

- Speed Estimation
- Motion Detection
- Breathing Rate Estimation



Set $v_i = 1 \text{ m/s}$, $f = 5.805 \text{ GHz}$
Theoretical Temporal ACFs



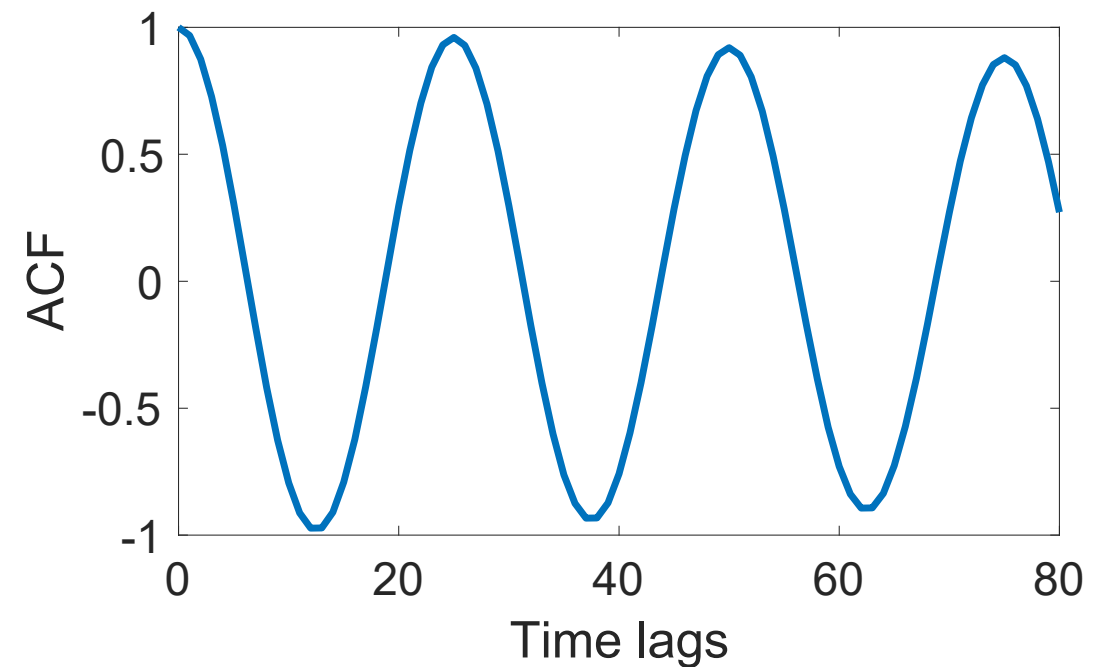
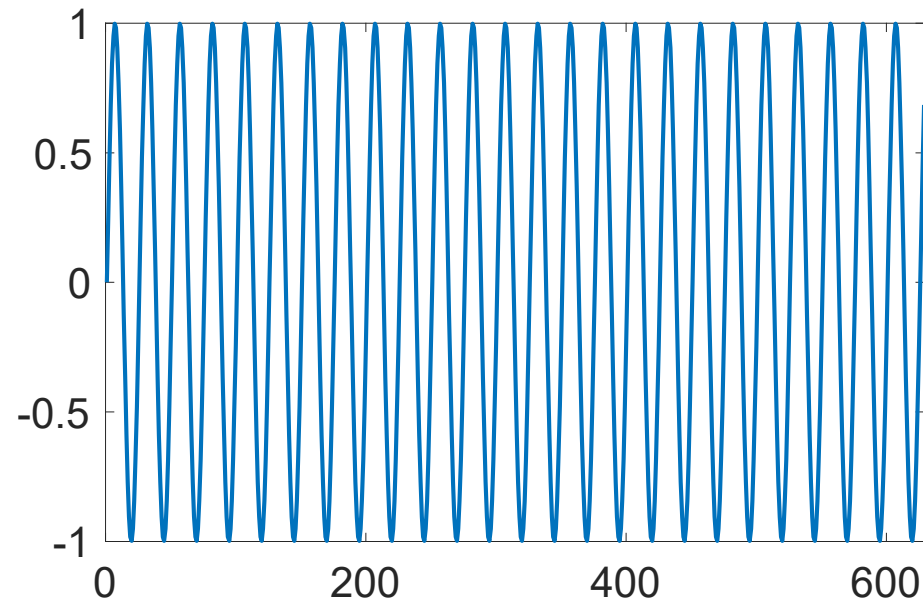
Review: What is ACF?

- Correlation of a signal (the same variables) with a delayed/lagged copy of itself
- We only consider a discrete time series $x[t]$, $t = 1, \dots, T$, the ACF is

$$r[k] = \frac{\sum_{t=k+1}^T (x[t] - \bar{x})(x[t-k] - \bar{x})}{\sum_{t=1}^T (x[t] - \bar{x})^2}$$

Review: What is ACF?

- Value range $[-1, 1]$
- A tool for finding period in the time domain $\leftarrow \text{autocorr}()$



Passive Speed Estimation



Applications



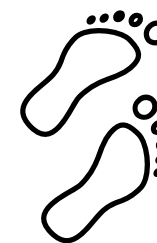
Fall Detection



Gait Recognition

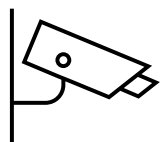


Activity Monitoring



Passive Tracking

Existing Solutions



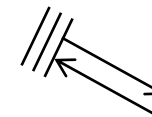
Camera

Very expensive
complicated calibration



Doppler Effect

Incomplete speed, LOS,
vulnerable to interferences



Time-of-Flight

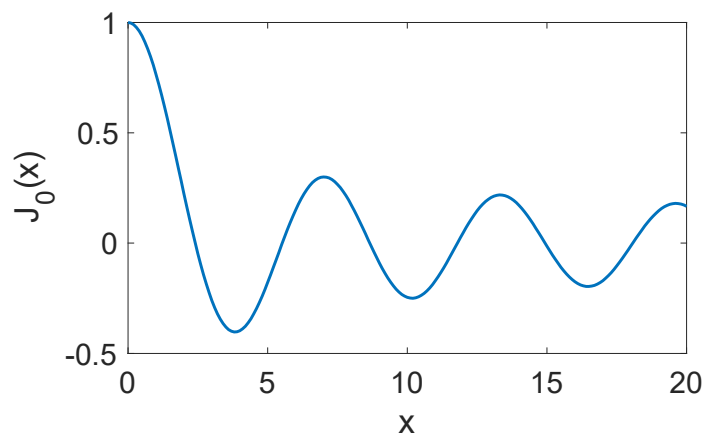
Inaccurate, LOS, barely
feasible on WiFi



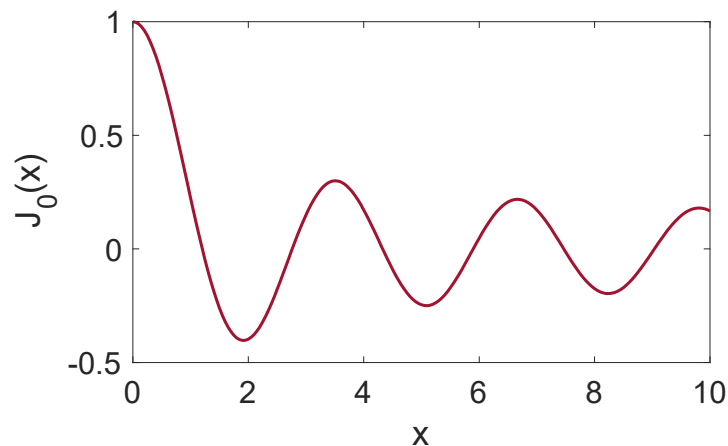
Speed Estimation with WiFi

- Estimate speed from patterns of $\rho_H(\tau, f)$

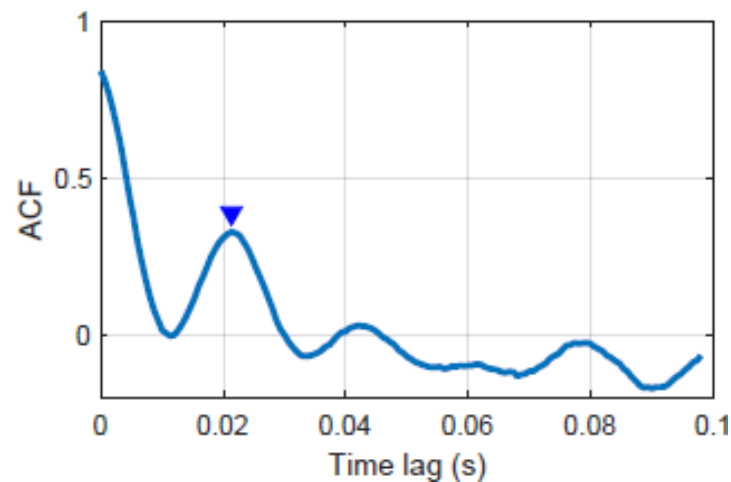
$$\rho_H(\tau, f) = \frac{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) + \sigma^2(f) \delta(\tau)}{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) + \sigma^2(f)} \underbrace{J_0(kv\tau)}_{0^{\text{th}}\text{-order Bessel function of the first kind}} \triangleq \alpha(f) J_0(kv\tau)$$



$J_0(x)$



$J_0(2x)$

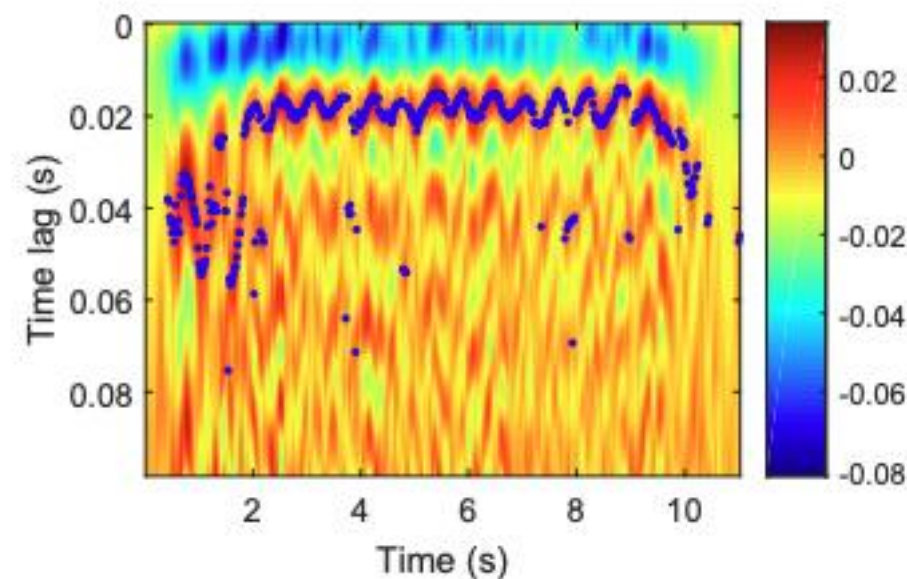
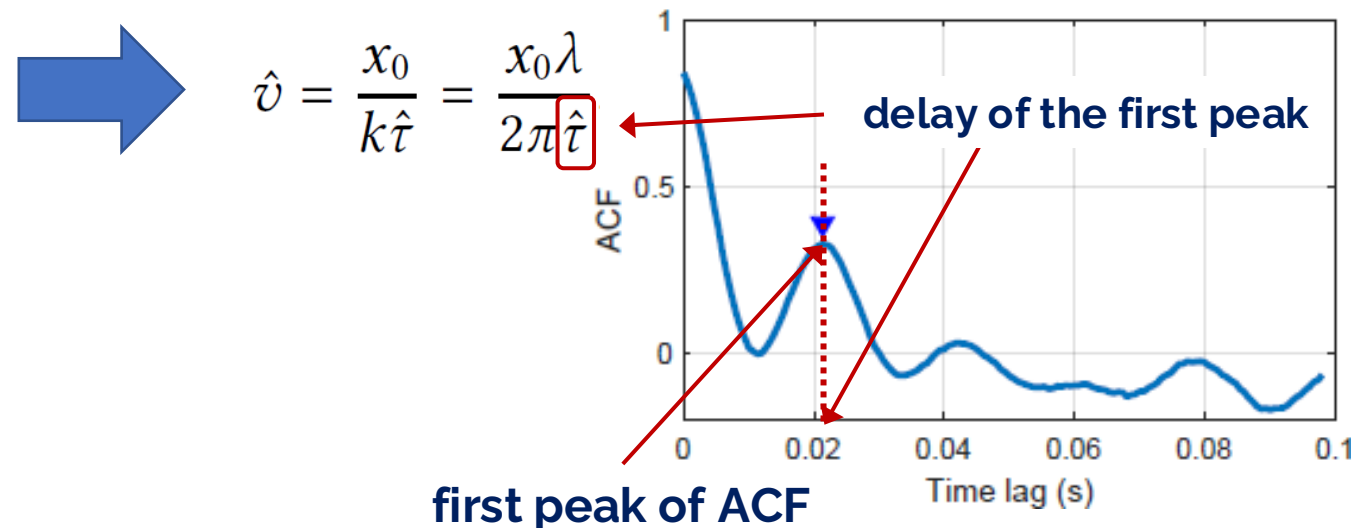
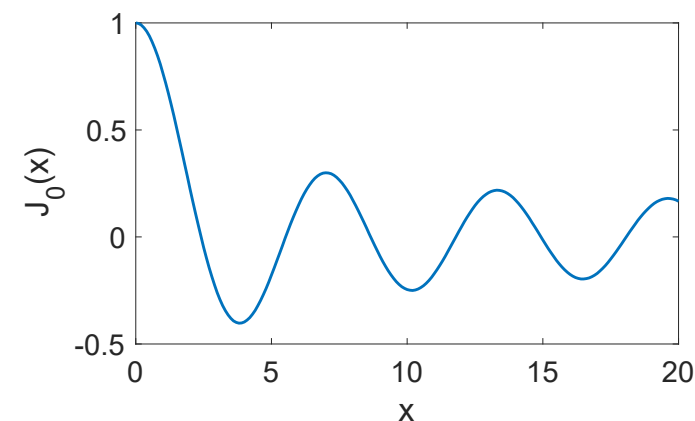


$\alpha J_0(ax)$

Speed Estimation with WiFi

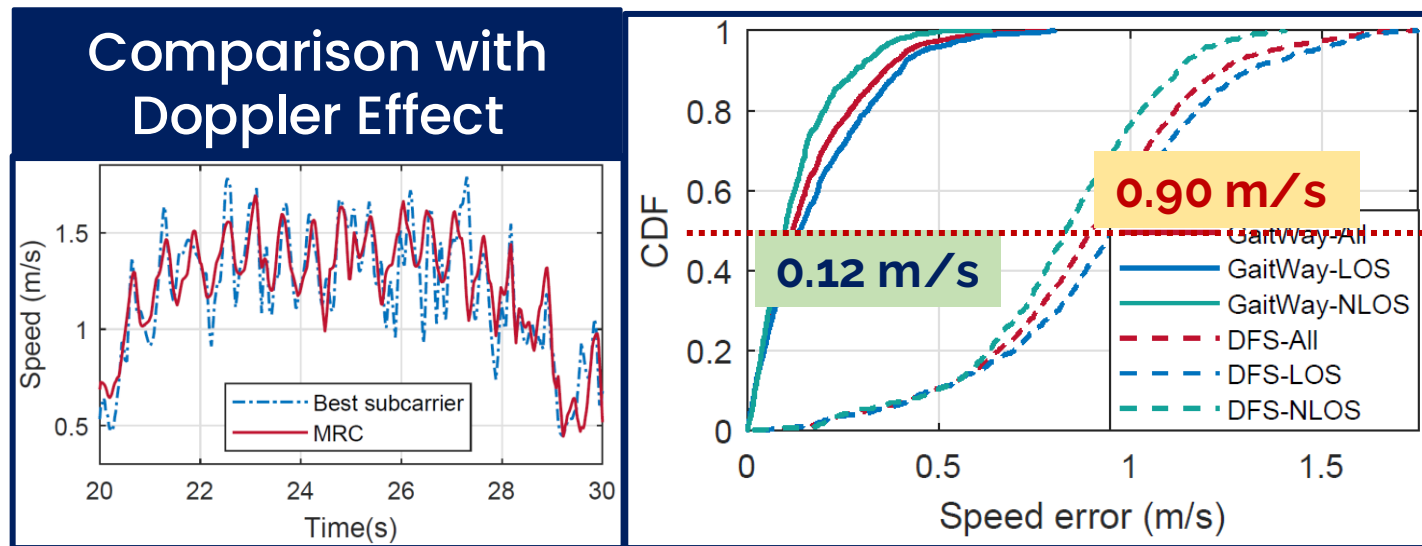
- Estimate speed from patterns of $\rho_H(\tau, f)$

$$\rho_H(\tau, f) = \frac{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) + \sigma^2(f) \delta(\tau)}{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) + \sigma^2(f)} \underbrace{J_0(kv\tau) \triangleq \alpha(f) J_0(kv\tau)}_{\text{0th-order Bessel function of the first kind}}$$



Speed Estimation with WiFi

- Comparison with Doppler-based methods



Advantages of statistical approaches:

- Works in NLOS
- Location and orientation independent
- Complete speed (thus more accurate)

Application: Fall Detection

Second leading cause

of accidental deaths worldwide
highest among adults over 60

646,000 deaths per year

80%+ in low/middle income areas

37.3M severe falls per year

Adults 65+ suffer the most

Fall detection is difficult!

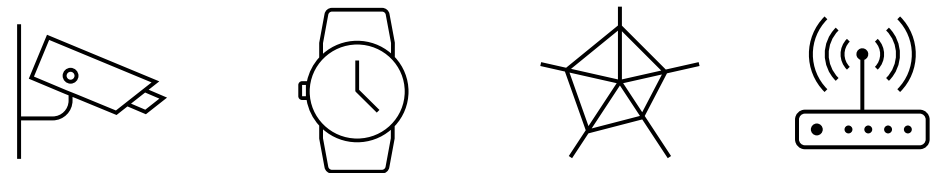
Cameras are **privacy-intrusive**

Wearables are **inaccurate and unfavorable**

UWB Radars have **limited coverage**

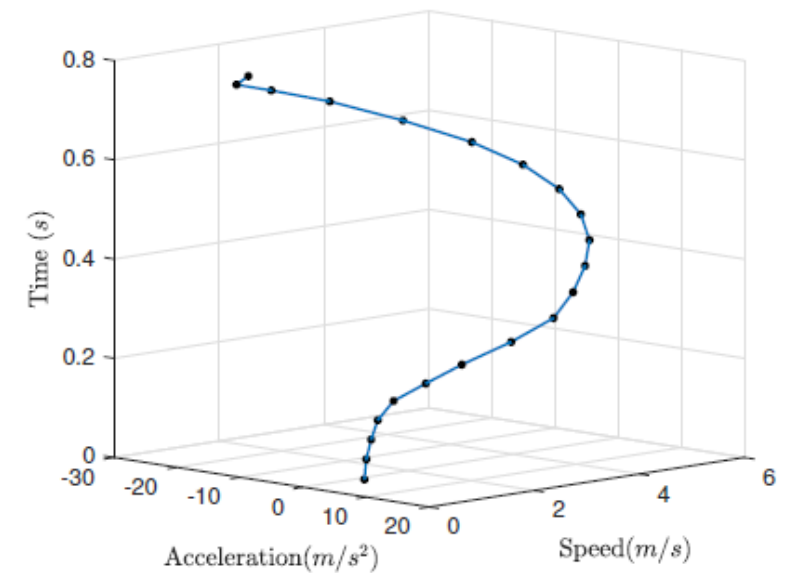
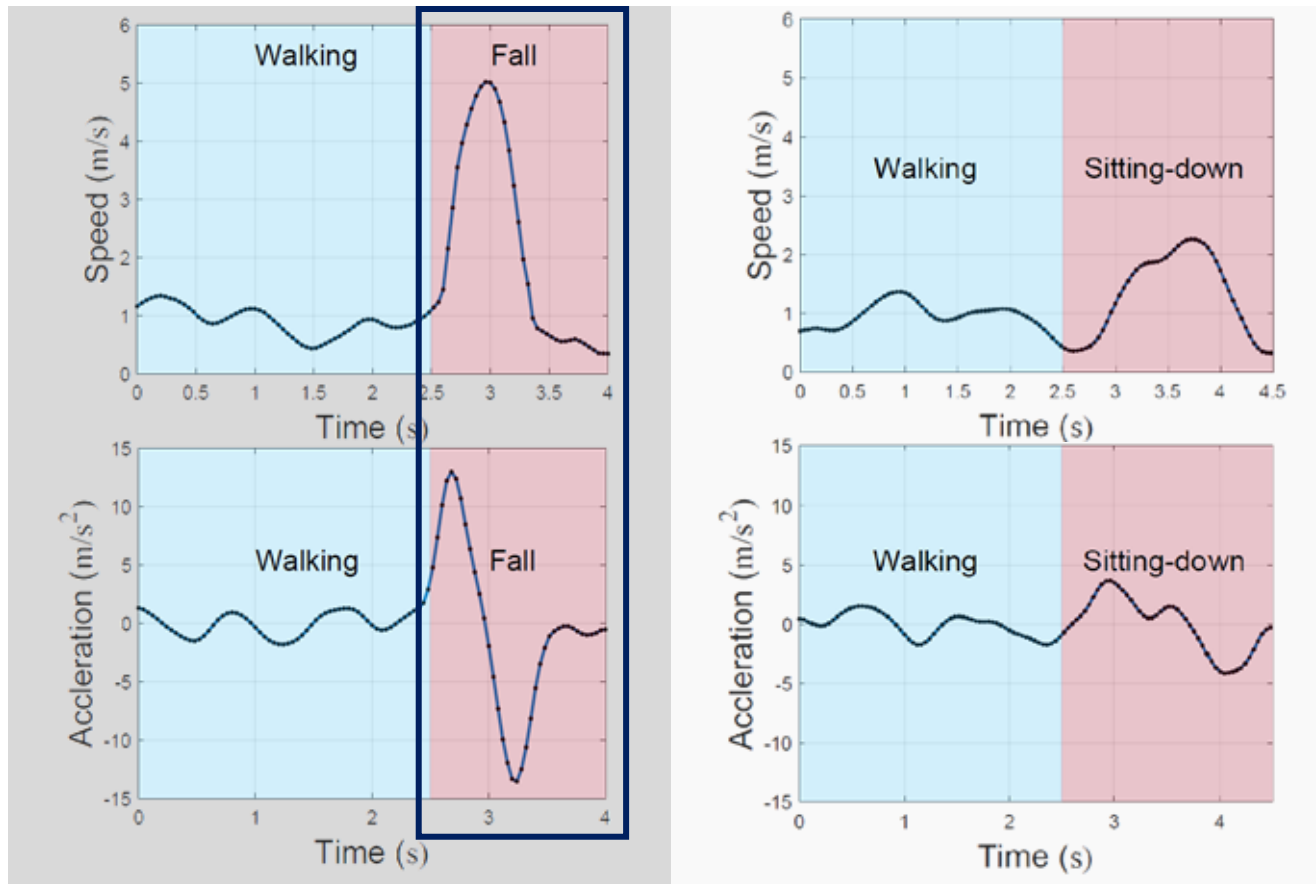
Existing WiFi methods **need training**

Low detection rate



Fall Detection by Speed

- Falls have distinct speed/acceleration patterns



Pattern Matching

Time series of speed & acceleration

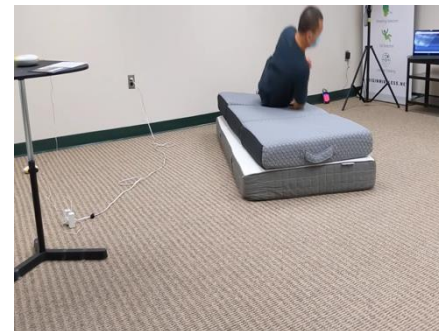
Evaluation

Offline Results

Activities	Features	Amount
Fall	LOS	518
	NLOS	328
Non-Fall	LOS	344
	NLOS	470

Methods	Features	Devices	Detection	False alarm
WiFall	Variance	3Tx/3Rx	87%	18%
RF-Fall	Spectrum	1Tx/2Rx	91%	11%
FallDeFi	Spectrum	2Tx/2Rx	95%/82%	15%/22%
DeFall	Speed	1Tx/1Rx	96.0%	1.5%

Real-time



138 / 140
falls
detected

Application: Gait Recognition

Gait as a Vital Sign

- Gait speed is termed as the **sixth vital sign**^[1]
- Indicative and predictive for general health and functional status



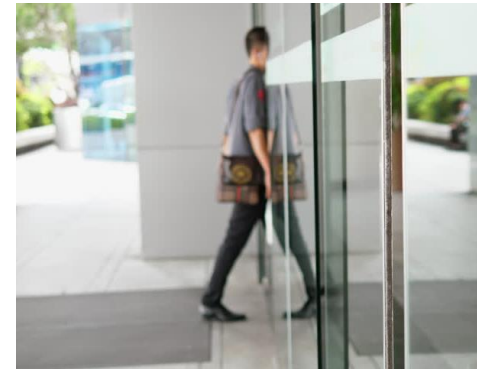
5 SURPRISING WAYS YOUR WALKING SPEED REFLECTS YOUR HEALTH



[1] Stacy Fritz and Michelle Lusardi. 2009. White Paper: Walking Speed: the Sixth Vital Sign. Journal of Geriatric Physical Therapy 32, 2(2009), 2.

Gait as a Biometric Marker

- Walking is a complex function coordinating every body system
- Difficult to impersonate others' walking patterns



GaitWay: Gait Speed Recognition

Speed Estimation



Stable Walking Extraction



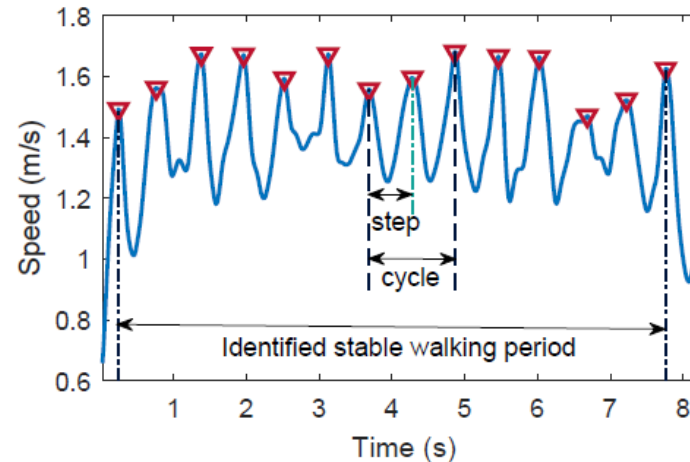
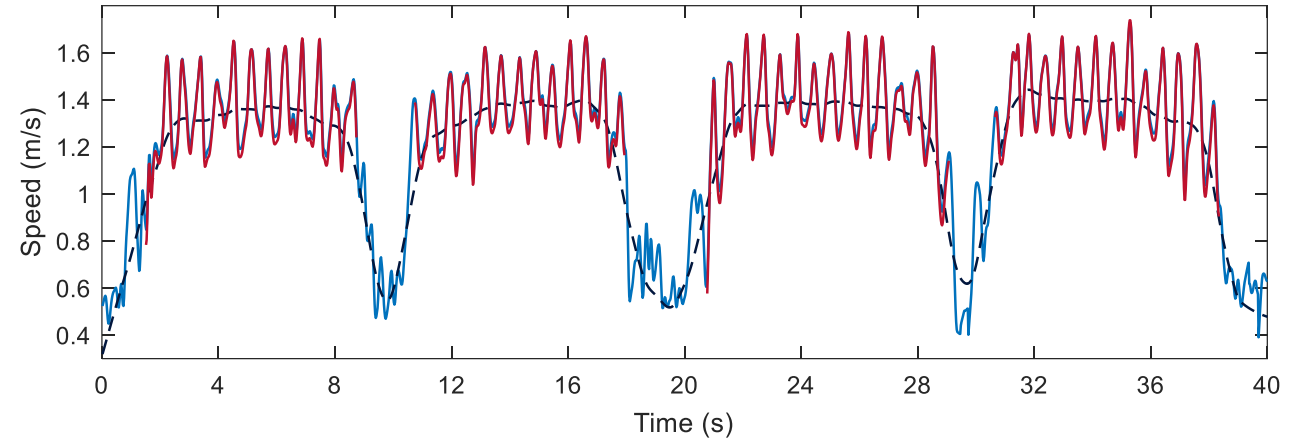
Gait Cycle Extraction



Gait Feature Extraction



Gait Recognition (SVM)



**Extract physically explainable
and environmentally
independent features**

symmetry, smoothness, variability,
periodicity, etc.

Whole-Home Motion Detection

Home Security

Intruder detection

Activity Monitoring

Occupancy detection

Home Automation

Presence detection

Infrared

LOS coverage

Cameras

False alarms
Privacy!

Vibration

Inaccurate
Deployment

Pressure

Confined coverage



Mission Impossible?

I (1996) →
← IV (2011)



Whole-Home Motion Sensing

- Recall ACF of $G(t, f)$:

$$\rho_G(\tau, f) = \frac{E_d^2(f)}{E_d^2(f) + \sigma^2(f)} \rho_\mu(\tau, f) + \frac{\sigma^2(f)}{E_d^2(f) + \sigma^2(f)} \delta(\tau)$$

If **motion**

$$\lim_{\tau \rightarrow 0} \rho_G(\tau, f) = \lim_{\tau \rightarrow 0} \frac{E_d^2(f)}{E_d^2(f) + \sigma^2(f)} \rho_\mu(\tau, f) > 0$$

If **no motion**

$$\lim_{\tau \rightarrow 0} \rho_G(\tau, f) = 0$$

Motion Statistic:

$$\hat{\phi}(f) \triangleq \hat{\rho}_G \left(\tau = \frac{1}{F_s}, f \right)$$



Photo courtesy: Origin Wireless

What's the statistical intuition?



Statistical Motion Detection

- When motion presents ($\mathcal{H}_1$ hypothesis):

$$\lim_{\tau \rightarrow 0} \rho_G(\tau, f) = \lim_{\tau \rightarrow 0} \frac{E_d^2(f)}{E_d^2(f) + \sigma^2(f)} \rho_\mu(\tau, f) > 0$$

- When no motion presents ($\mathcal{H}_0$ hypothesis):

$$\lim_{\tau \rightarrow 0} \rho_G(\tau, f) = 0$$

- Define ***motion statistic*** on subcarrier f as:

$$\hat{\phi}(f) \triangleq \hat{\rho}_G \left(\tau = \frac{1}{F_s}, f \right)$$

Statistical Motion Detection

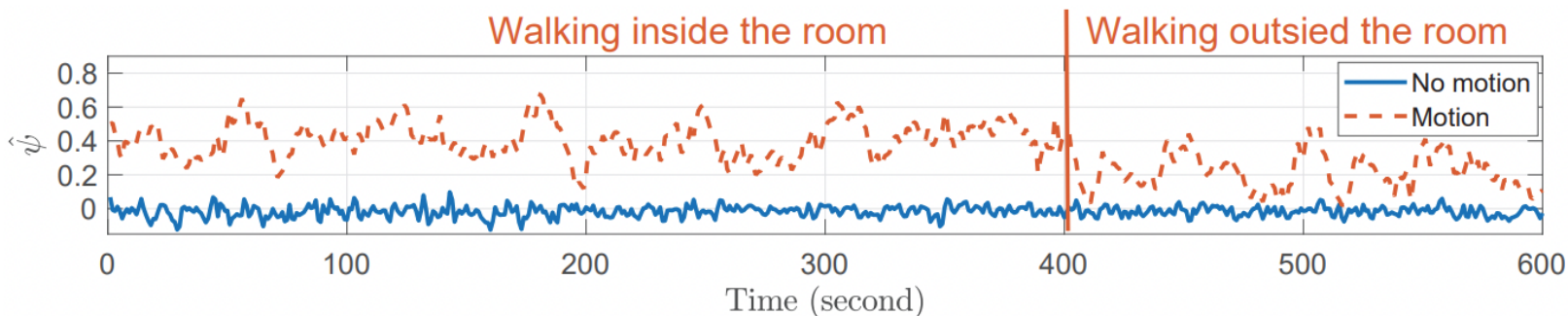
- Statistical Distributions of the *Motion Statistic*

$$\mathcal{H}_0 : \hat{\phi}(f) \sim \mathcal{N}(-\frac{1}{T}, \frac{1}{T}), \forall f \in \mathcal{F}, \text{ where } T \text{ is the number of samples.}$$

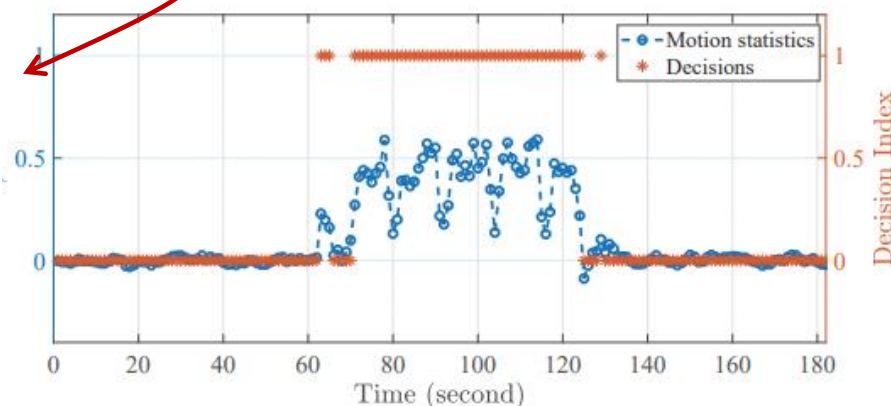
$$\mathcal{H}_1 : \hat{\phi}(f) \rightarrow \frac{E_d^2(f) + E_s^2(f)}{E_d^2(f) + E_s^2(f) + \sigma^2(f)} > 0, \forall f \in \mathcal{F}$$

- Detection Rule (Average over all subcarriers)

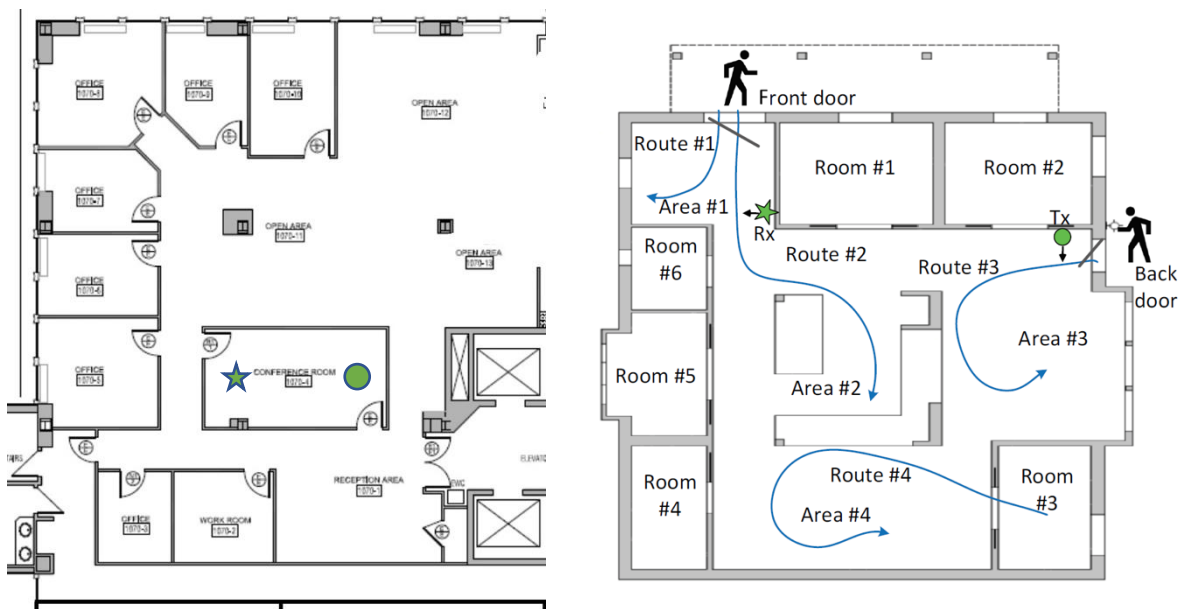
$$\text{Reject } \mathcal{H}_0 \text{ iff } \hat{\psi} \triangleq \frac{1}{F} \sum_{f \in \mathcal{F}} \hat{\phi}(f) > \eta$$



Motion indicator &
Motion strength indicator



Experiments

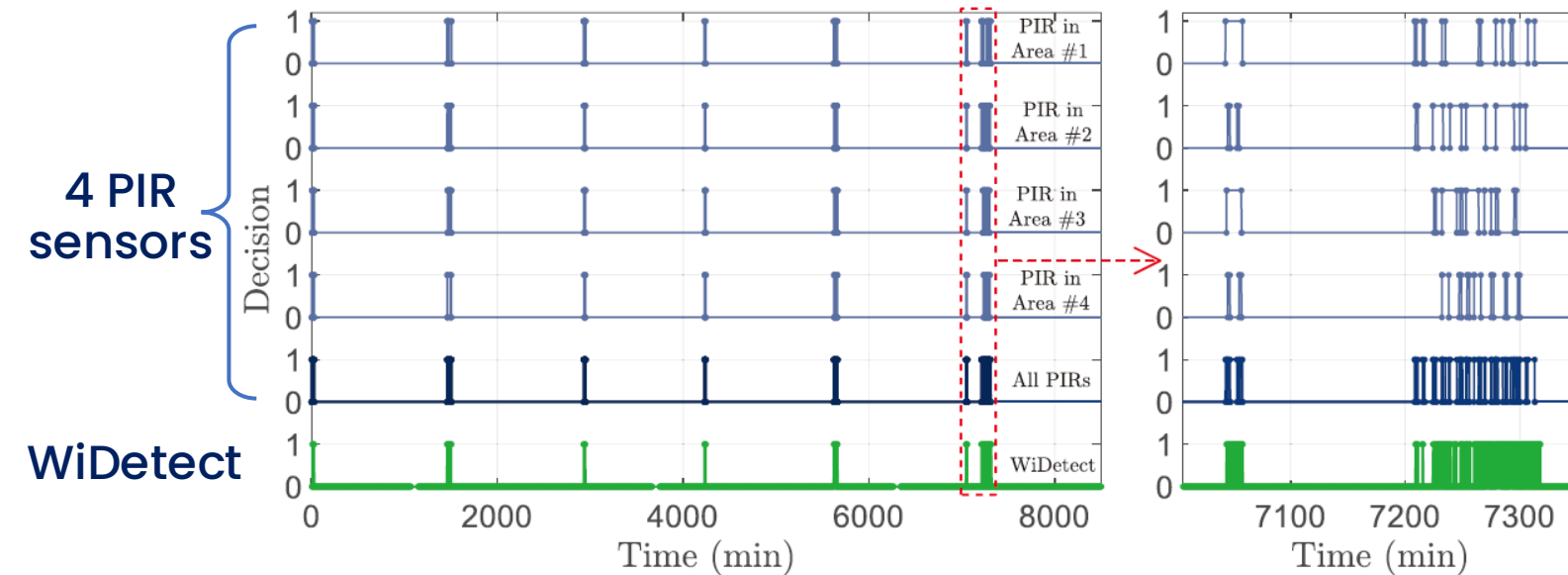


3x3 MIMO WiFi, **5.8GHz** channel, **30-180Hz** rate

- 1) **Office**: parameter study
- 2) Single **house**: coverage and PIR comparison
- 3) 1B1B **apartment**: 1-month deployment

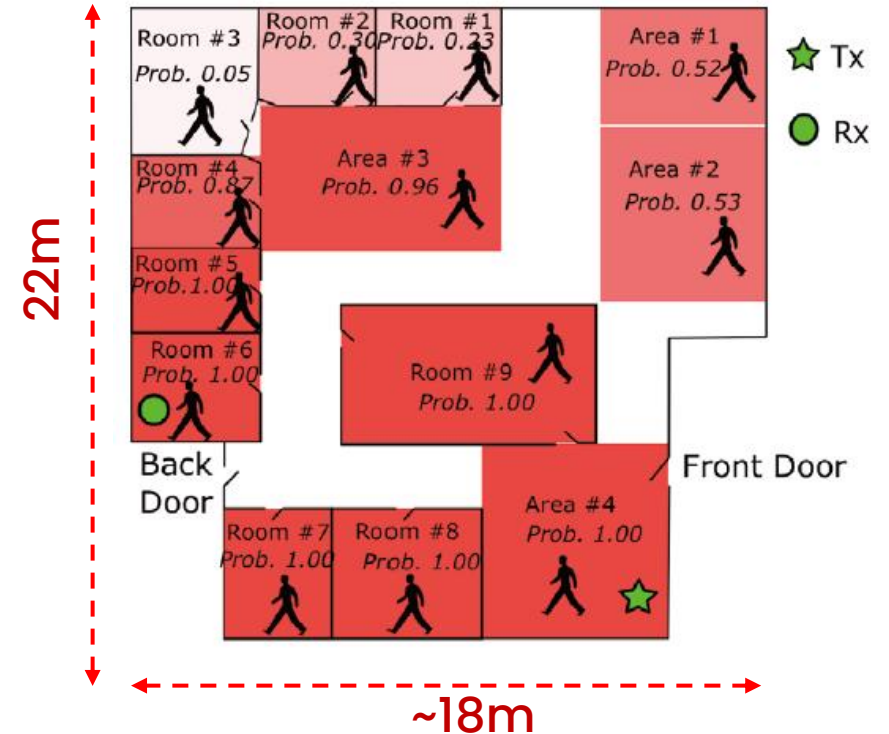
Methods	Claimed FN	Claimed FP	Calibration
RASID	4.7%	3.8%	Yes
PILOT	10.0%	10.0%	Yes
E-Eyes	10.0%	1.0%	Yes
Omni-PHD	8.0%	7.0%	Yes
DeMan	5.9%	1.5%	Yes
CARM	2.0%	1.4 per hour	Yes
SIED	6.4%	2.0%	Yes
FreeSense	1.4%	0.5%	Yes
WiDetect	0.3%	0.0%	NO

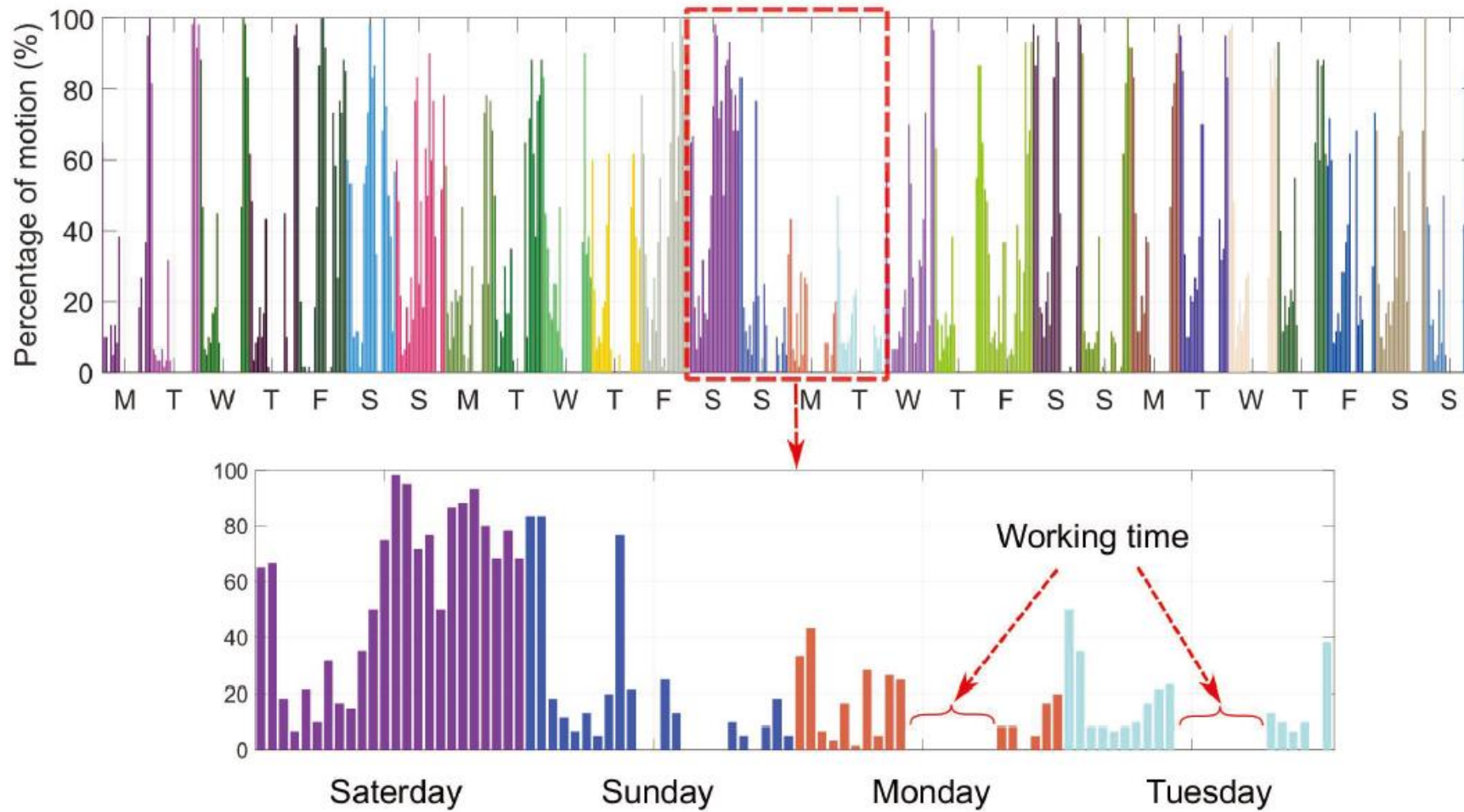
Experiments: Long-term & Coverage



Mission Impossible!

A false alarm of 10^{-6} for a total of 4 weeks of data collected in 4 facilities!





Real-World Case Study

Activity of Daily Living Monitoring

Sleep Monitoring

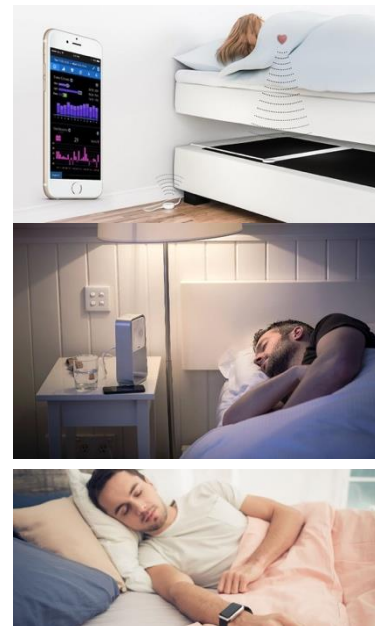


Sleep is vital
Mentally & physically

10% suffer from chronic insomnia
1/3 of Americans short of sleep



Polysomnography(PSG)
\$1300/night, too invasive



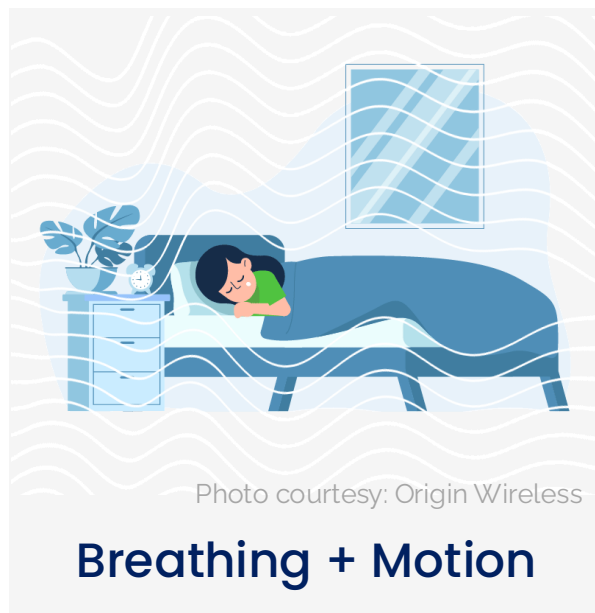
Pressure mat
inaccurate

UWB Radar
Inaccurate
short range

Wearables
Inaccurate

SMARS

Sleep Monitoring via
Ambient Radio Signals



Breathing Signal

- Measurement of Breathing Signal Using CSI
- Breathing is periodic

$$|H(t, f)|^2 = \underbrace{g(f)}_{\text{channel gain}} \underbrace{b(t - \Delta t_f)}_{\text{breathing signal}}$$

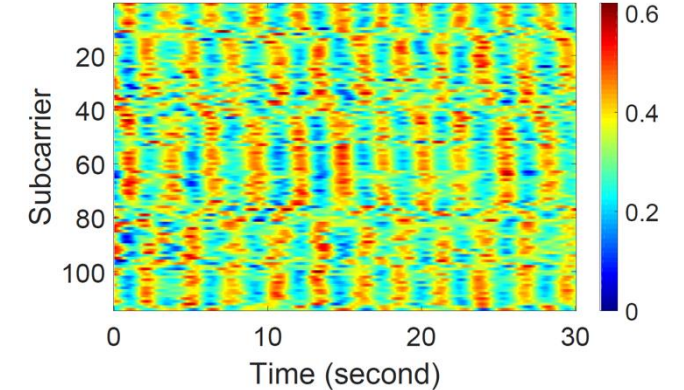
where $k(f) \triangleq \frac{g^2(f)}{g^2(f) + \sigma^2(f)}$, and $n(\tau, f) \sim \mathcal{N}(0, \frac{1}{N})$, $\forall f, \forall \tau$

$$\hat{\rho}_G(\tau, f) \approx k(f) \underbrace{\rho_b(\tau)}_{\text{ACF of } b(t)} + n(\tau, f),$$

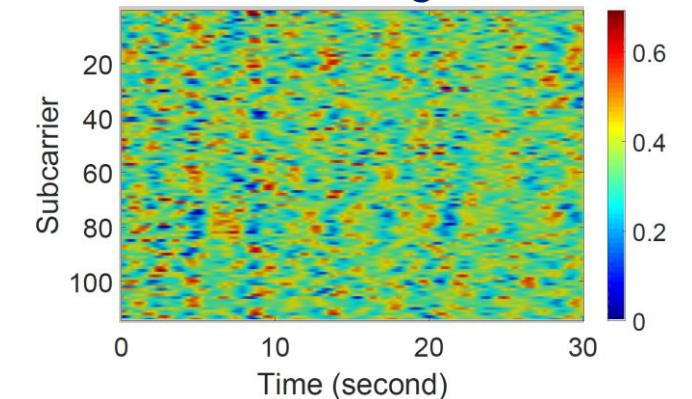


Photo courtesy: The Big Bang Theory

Line-Of-Sight (LOS)



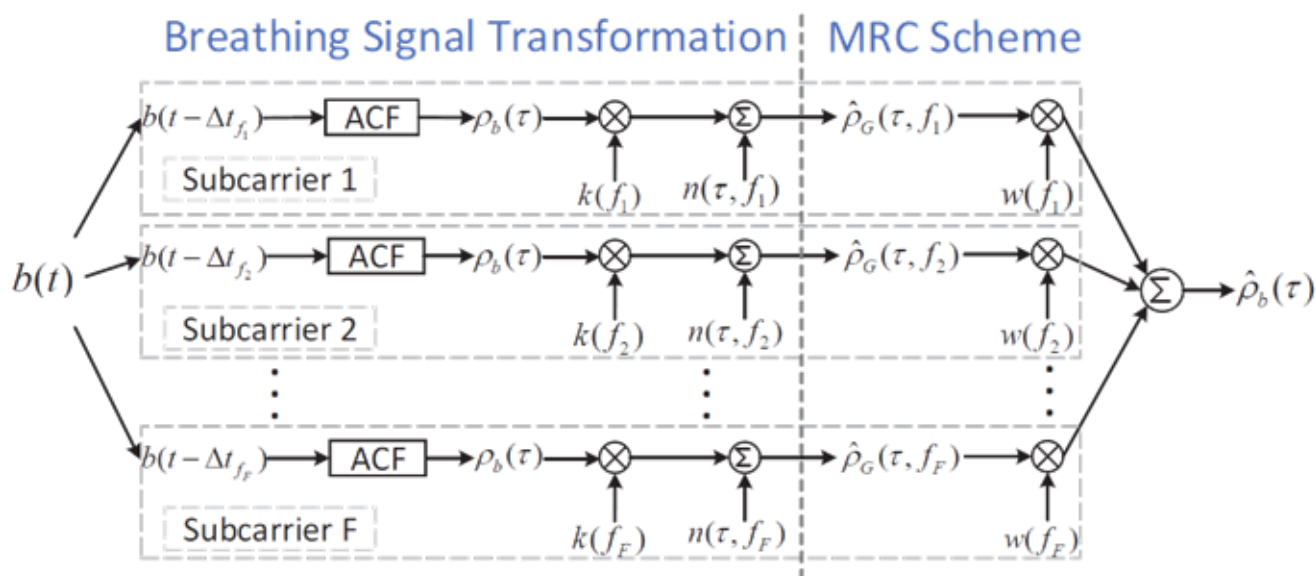
Non-Line-Of-Sight (NLOS)



Boosting Weak Breathing Signals

- Maximize Breathing Signal by Maximal Ratio Combining (MRC)
- Frequency diversity: frequency-selective fading

Maximal Ratio Combining (MRC)

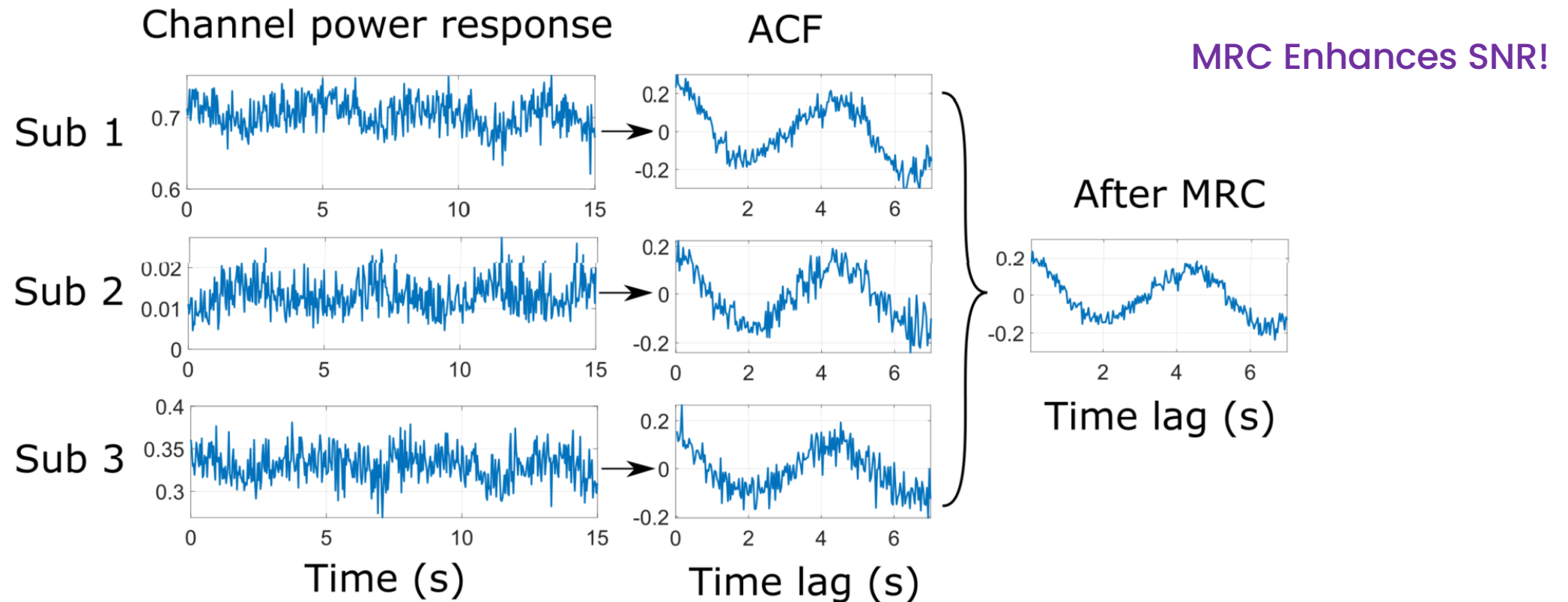


$$w(f) = \hat{k}(f) = \hat{\rho}_G \left(\tau = \frac{1}{F_s}, f \right)$$

$$\begin{aligned} \hat{\rho}_b(\tau) &= \sum_{f \in \mathcal{F}} w^*(f) \hat{\rho}_G(\tau, f) \\ &= \sum_{f \in \mathcal{F}} \hat{\rho}_G(\tau = 1/F_s, f) \hat{\rho}_G(\tau, f). \end{aligned}$$

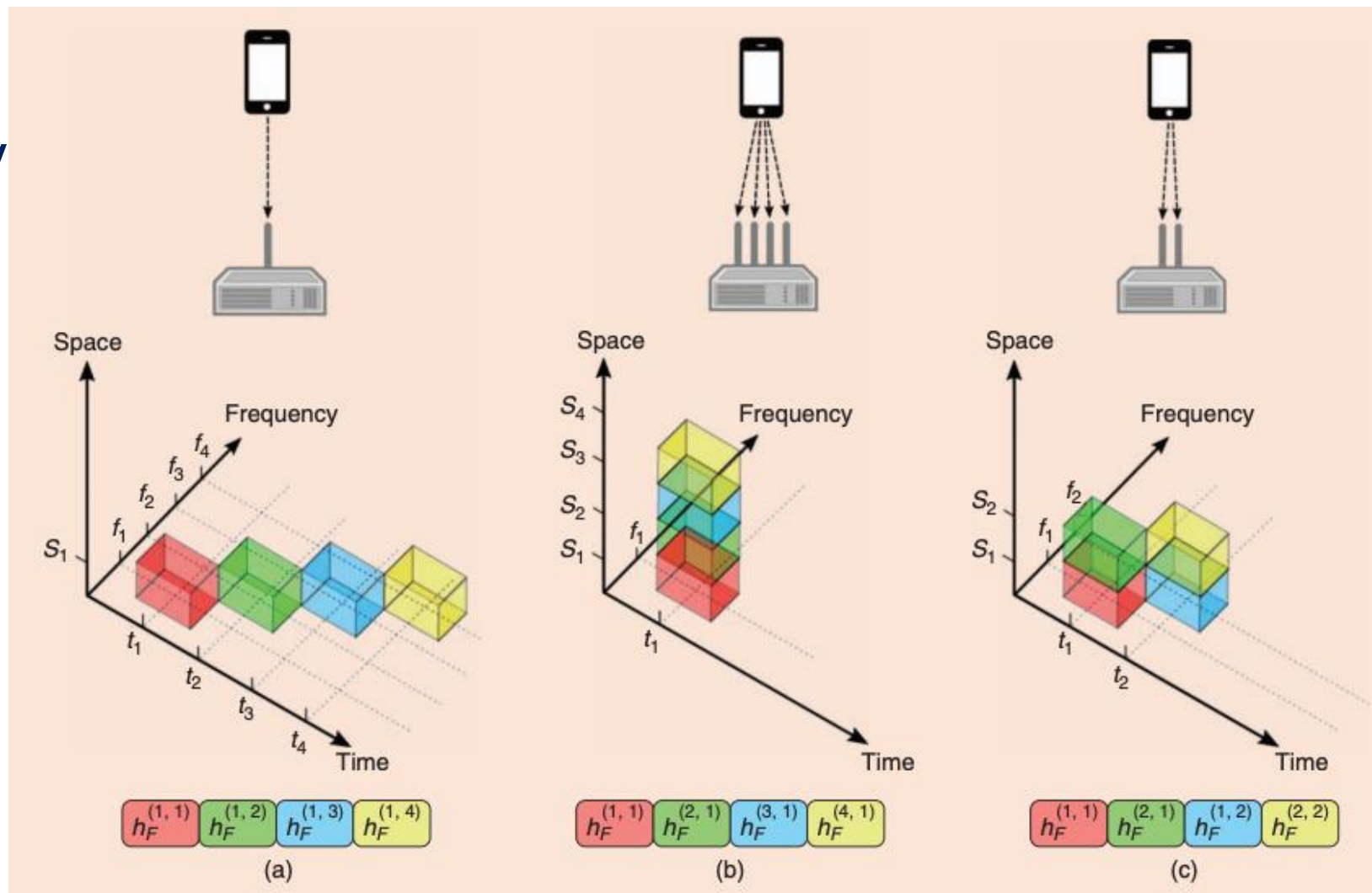
Boosting Weak Breathing Signals

- Breathing Signal Maximization Using MRC



Diversity

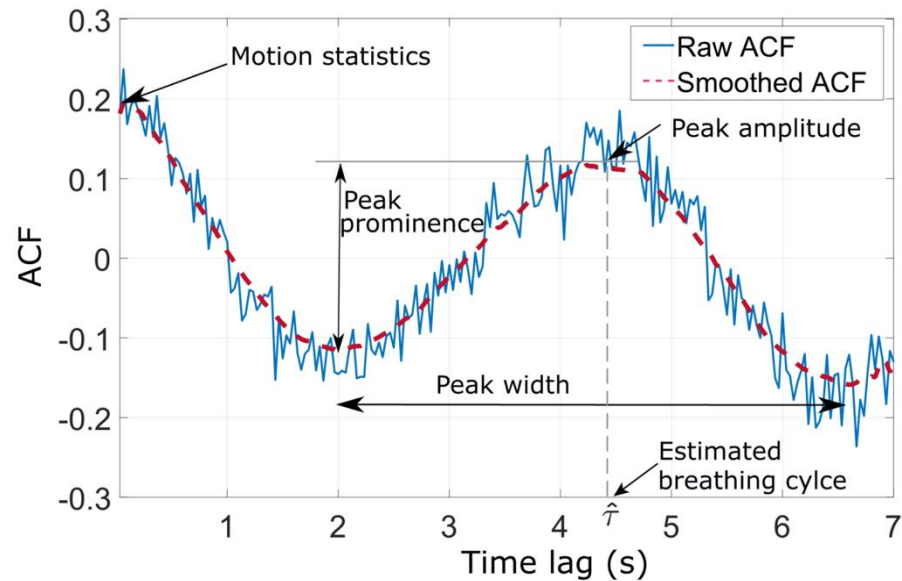
- Frequency diversity
- Space diversity
- Time diversity



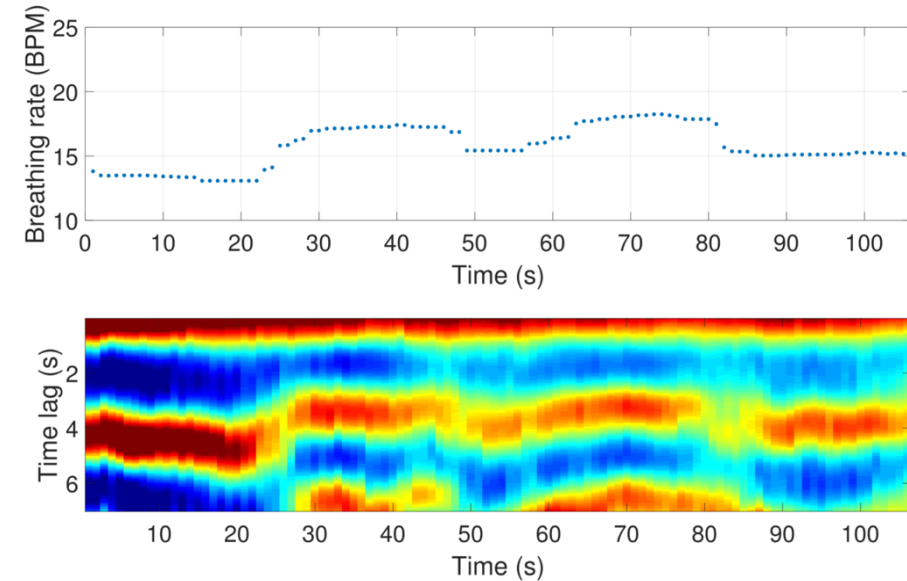
Wang, Beibei, et al. "The promise of radio analytics: A future paradigm of wireless positioning, tracking, and sensing." IEEE Signal Processing Magazine 35.3 (2018): 59-80.

Instantaneous Breathing Estimation

- Features for Breathing Estimation



- Example of Breathing Estimation



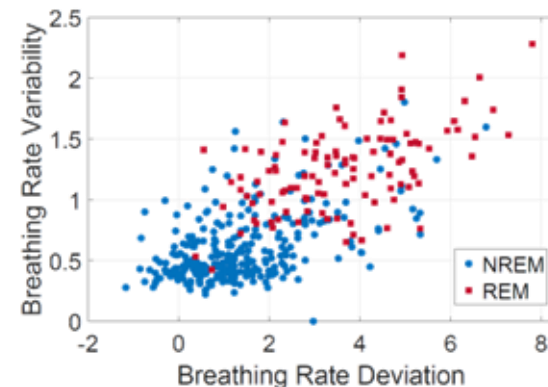
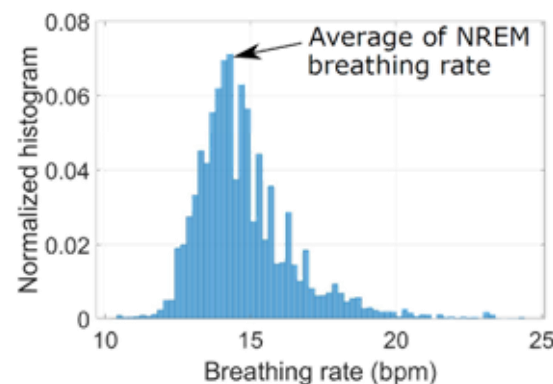
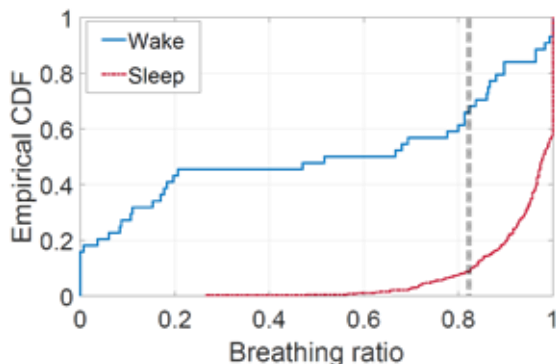
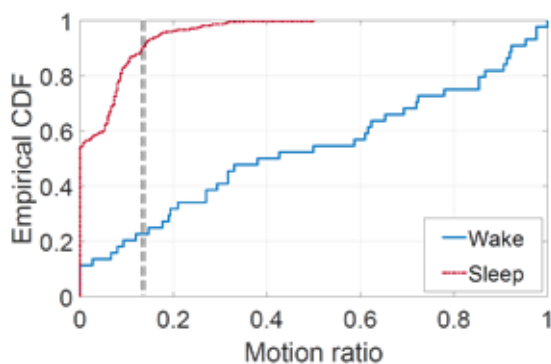
Motion & Breathing for Sleep Staging

- Sleep/Wake Classification

- **Motion ratio**: the percentage of time when the motion is detected;
- **Breathing ratio**: the percentage of time when breathing signal is detected.

- REM/NREM Classification

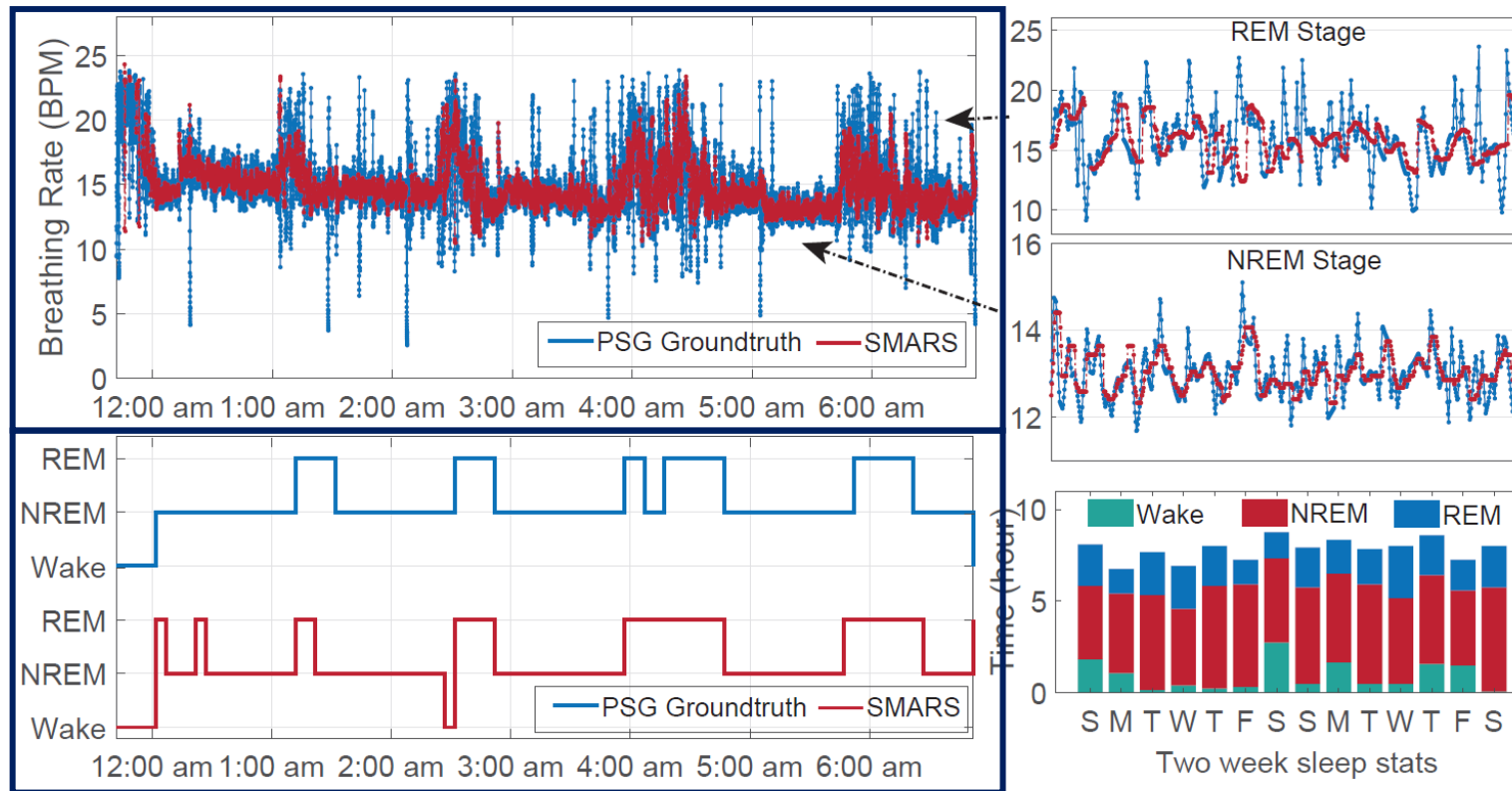
- **Breathing rate variability**: the variance of breathing estimates;
- **Breathing rate deviation**: the deviation of breathing estimates from average.



Wake/Sleep Classification

Rapid Eye Movement (REM)/Non-REM Staging

Sleep Monitoring Results



A Unified Framework

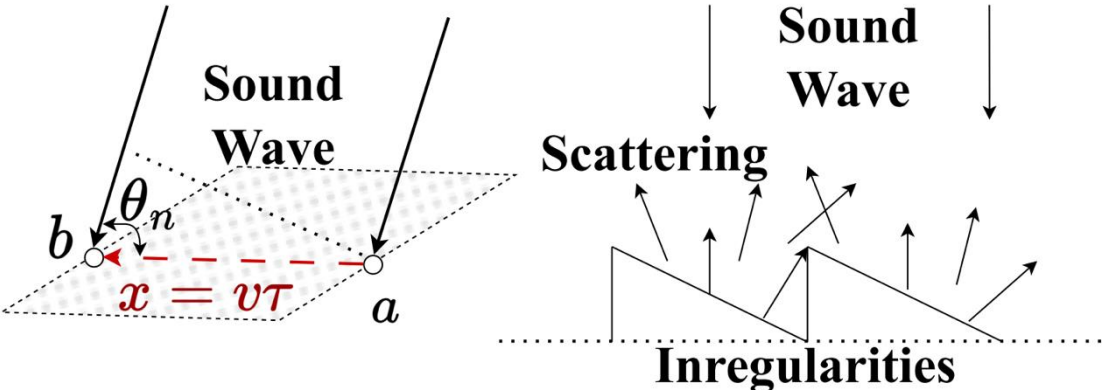
- All based on ACF
 - motion, speed, breathing
- A unified pipeline
 - A time series of CSI
 - Consider a specific window and calculate the ACF
 - Slide the window and continue to calculate the ACF
 - Find
 - The value of the first sample for motion
 - The first peak in the ACF for breathing or speed

CSI Tools

- Intel 5300 NIC CSI Tool (2012, the first one)
 - Intel WiFi Wireless Link 5300 802.11n
 - 30 subcarriers
- Atheros CSI Tool (2015)
 - Qualcomm Atheros series
- Nexmon (2017)
 - Broadcom Wi-Fi Chips, can support measurement on phones!
- PicoScenes (2019)
 - 802.11ac/ax
 - Qualcomm Atheros AR9300 (QCA9300), Intel Wireless Link 5300 (IWL5300), Intel AX200 and Intel AX210.
- ESP32 CSI Tool
 - The only one commercially supported

Even holds with sound signals!

- Do the same/similar properties hold for acoustics?
 - WiFi: Electromagnetic waves
 - Sound: Mechanical waves
- Sound Diffusion Model in Room Acoustics



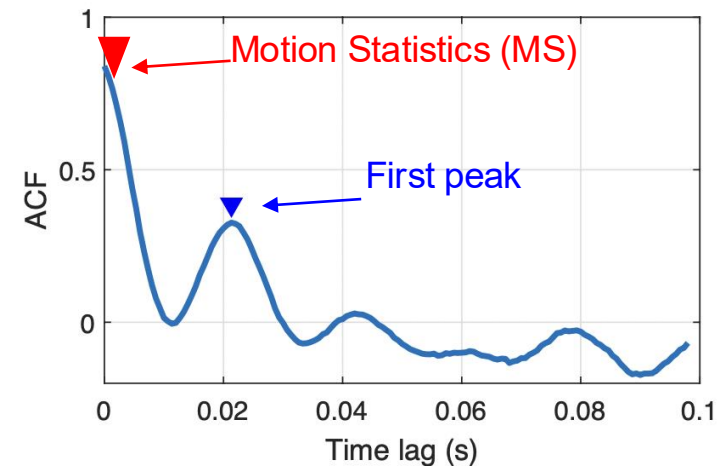
$$\psi_p(x) = \frac{1}{2\pi} \int_0^{2\pi} \psi(x, \theta) d\theta = J_0(kx),$$

$$J_0(x) = \frac{1}{2\pi} \int_0^{2\pi} \cos(x \cos \theta) d\theta$$

Statistical Acoustic Sensing (SAS)

ACF of CSI resembles 0th-order Bessel function of its first kind.

A function of speed v , independent of moving directions and locations.



Acoustic Channel State Information (CSI)

$$H(f, t) = \sum_{i \in R_D} H_i(f, t) + \sum_{j \in R_S} H_j(f, t) + N(f, t)$$

Autocorrelation Function (ACF) of CSI

$$\rho(f, \tau) = \frac{\sum_{i \in R_D} 2\pi\sigma_i^2(f) + \sigma_N^2(f)\delta(\tau)}{\sum_{i \in R_D} 2\pi\sigma_i^2(f) + \sigma_N^2(f)} J_0(kv\tau)$$

$$\triangleq g(f)J_0(kv\tau), \quad \tau \neq 0$$



Motion: $g(f) = \lim_{\tau \rightarrow 0} \rho(f, \tau) = \tilde{\rho}(f, \tau = 1/F_s)$

The value of the first sample in ACF

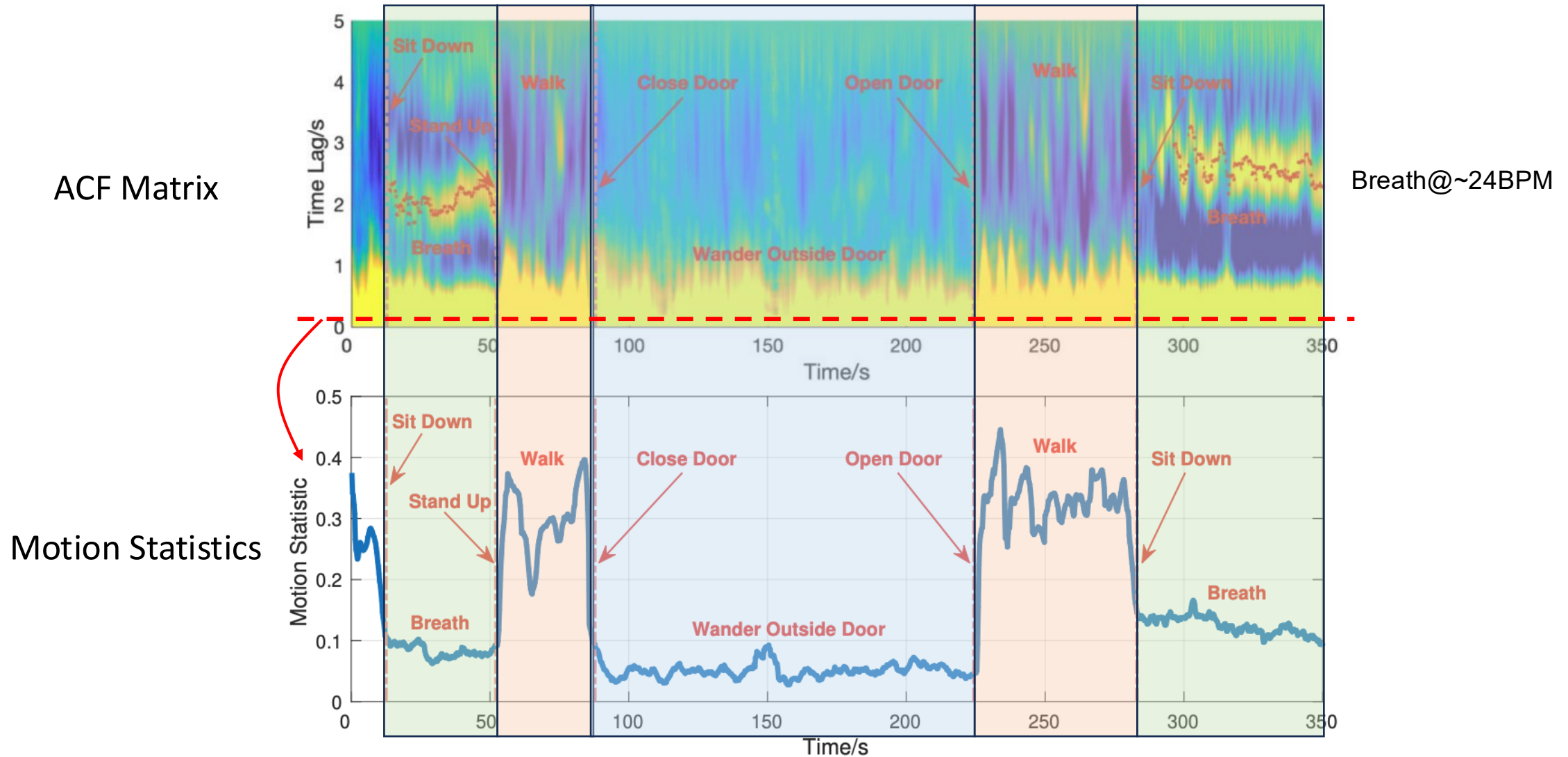
Breath: $f_{breath} = 60/\tau_b$

Find the delay of the first peak in ACF

Speed: $v = \frac{x_0}{k\tau_s} = \frac{x_0\lambda}{2\pi\tau_s}$

Find the delay of the first peak in ACF

Statistical Acoustic Sensing (SAS)



VeCare: In-Cabin AI with SAS

VeCare: A early attempt for Child Presence Detection using commodity in-car audio devices

Child Presence Detection

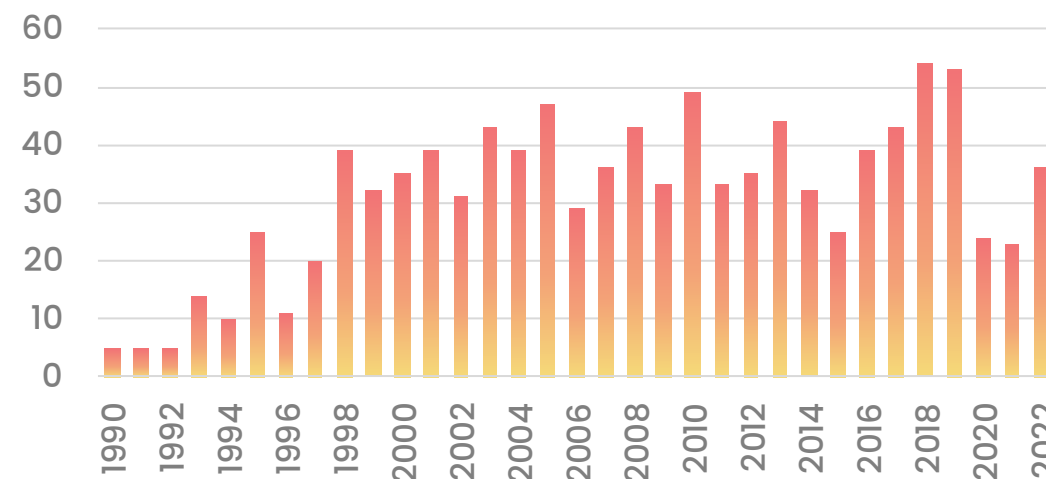
Detect unattended children left in a car to avoid heatstroke deaths

Child Vehicular Heatstroke Deaths

39 (one every 9 days) deaths per year,

1000+ children have died since 1990 in US *

*Data source: KidsAndCars.org



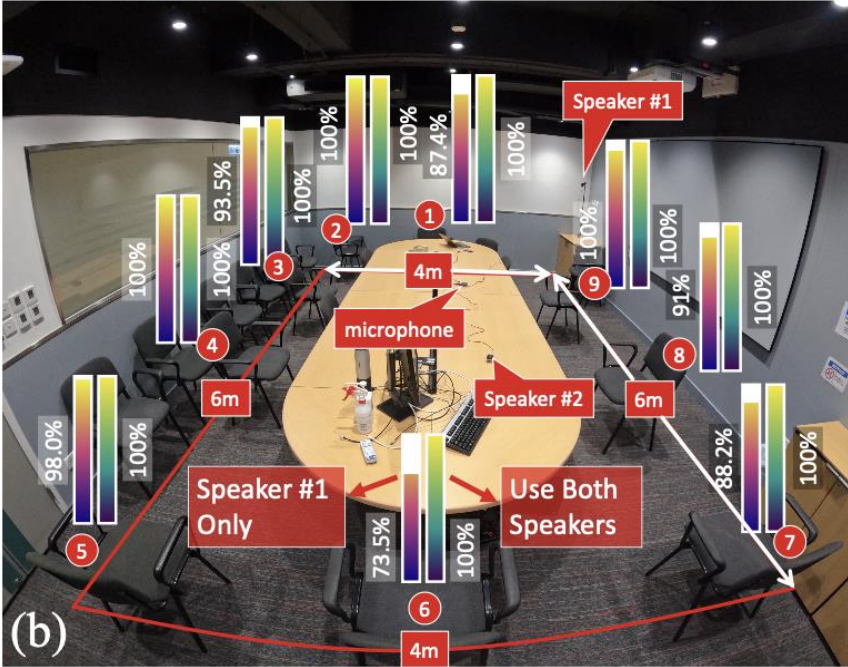
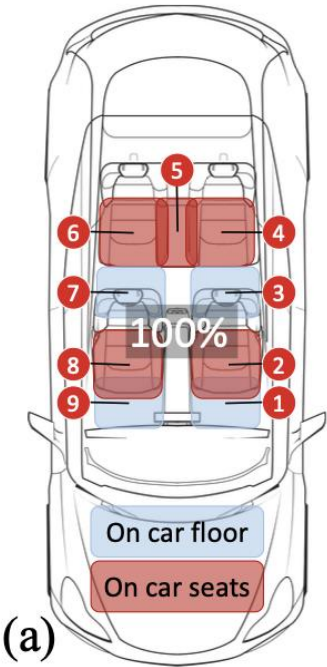
VeCare: In-Cabin AI with SAS

CPD with no blind spot!



NO MORE
Hot Car Deaths!

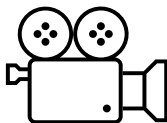
VeCare: SAS for Child Presence Detection



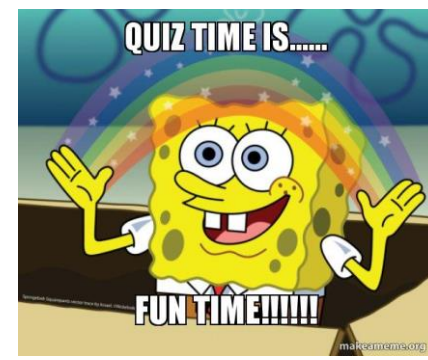
Respiration Monitoring



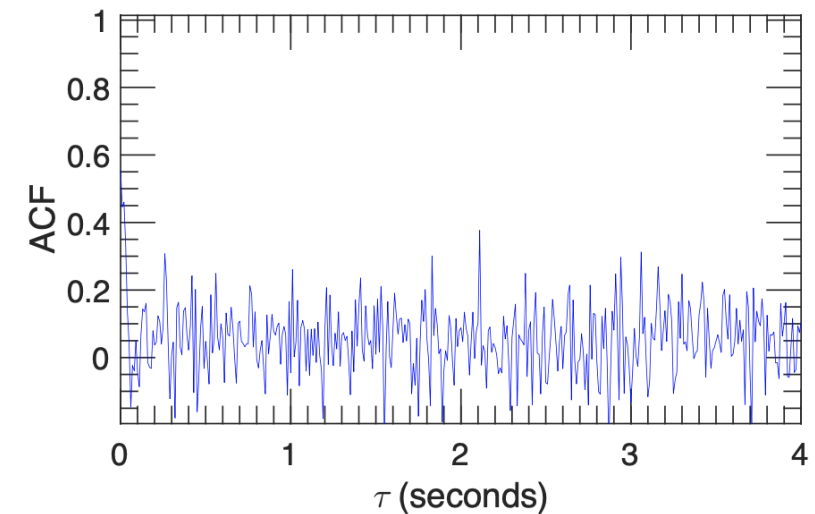
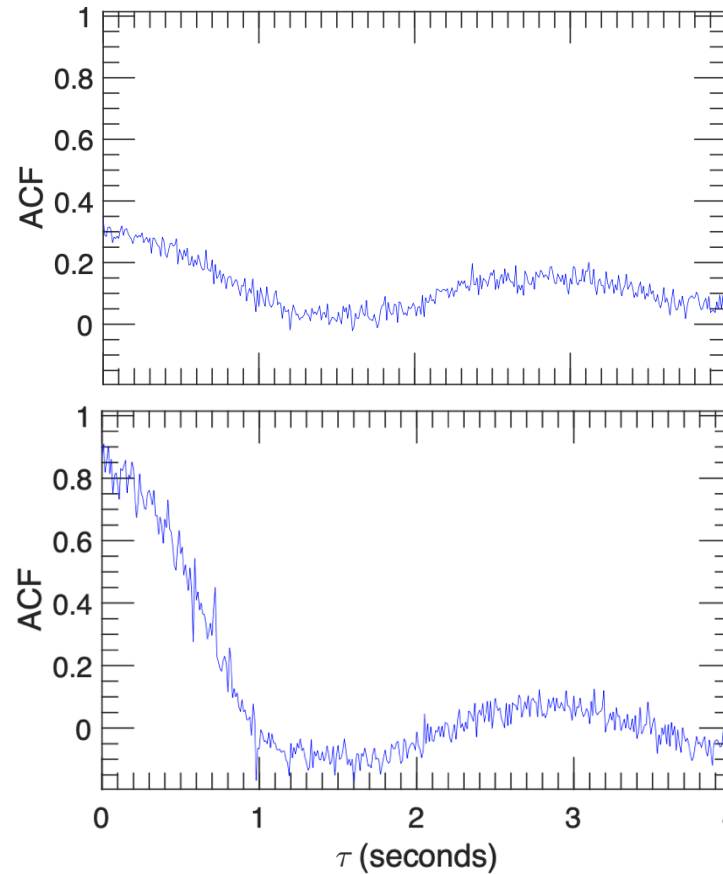
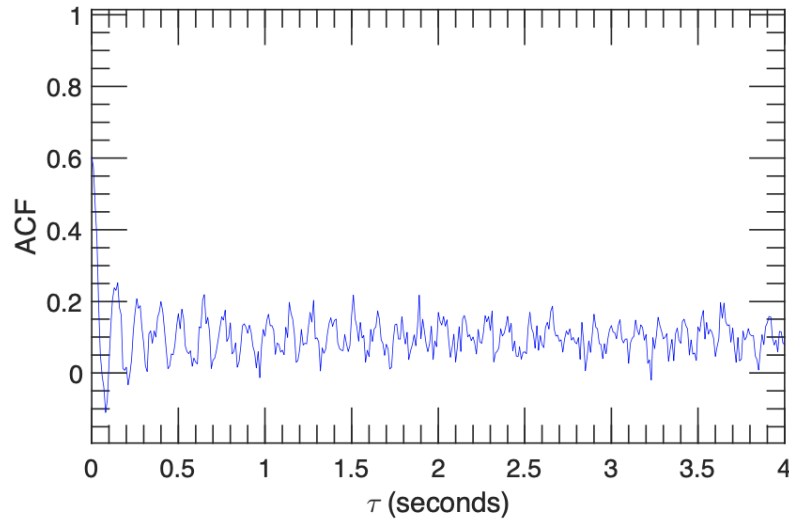
Motion Low



Quiz



- What is (most likely) going on given the following ACF of CSI?



Questions

- What windows should we use for calculation?
- How to determine whether it is breathing or speed?
- How much computation is needed?



More Questions

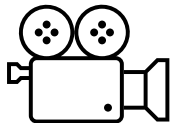
- Why not FFT?
 - FFT is a (more) common tool for finding frequency/periodicity.
- Why not Deep Learning?
 - DL is now a popular tool for finding X.
 - →→ Lecture on Deep Wireless Sensing
- Limitations of (current) WiFi sensing?



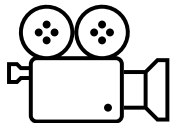
Contents

- Multipath Effect
 - Reflection Model
 - Scattering Model
- Geometrical Approaches
- Statistical Approaches
 - Speed Estimation
 - Motion Detection
 - Breathing Rate Estimation
- More Applications

Person Re-ID



WiFi Imaging and Pose



[MobiCom 2020] Towards 3D Human Pose Construction Using WiFi

WiFi Imaging

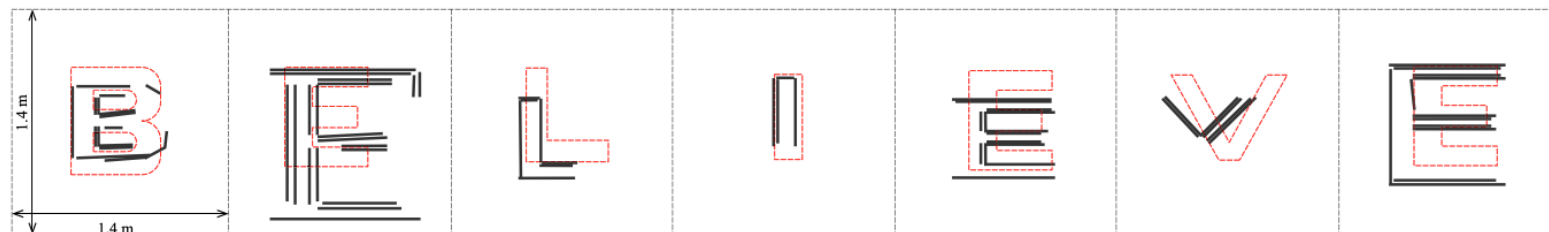
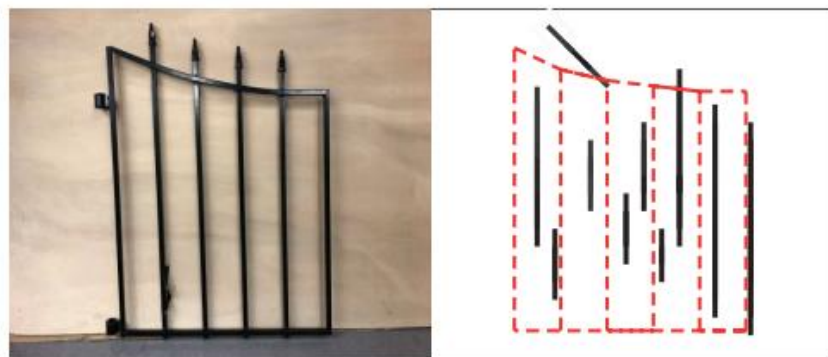


Figure 11: Through-wall reading: Wifract enabling WiFi to image and read the letters of the word "BELIEVE" behind the wall of Area 3. The dashed lines represent the ground-truth while the solid lines represent our image.

Garden fence



Microwave oven

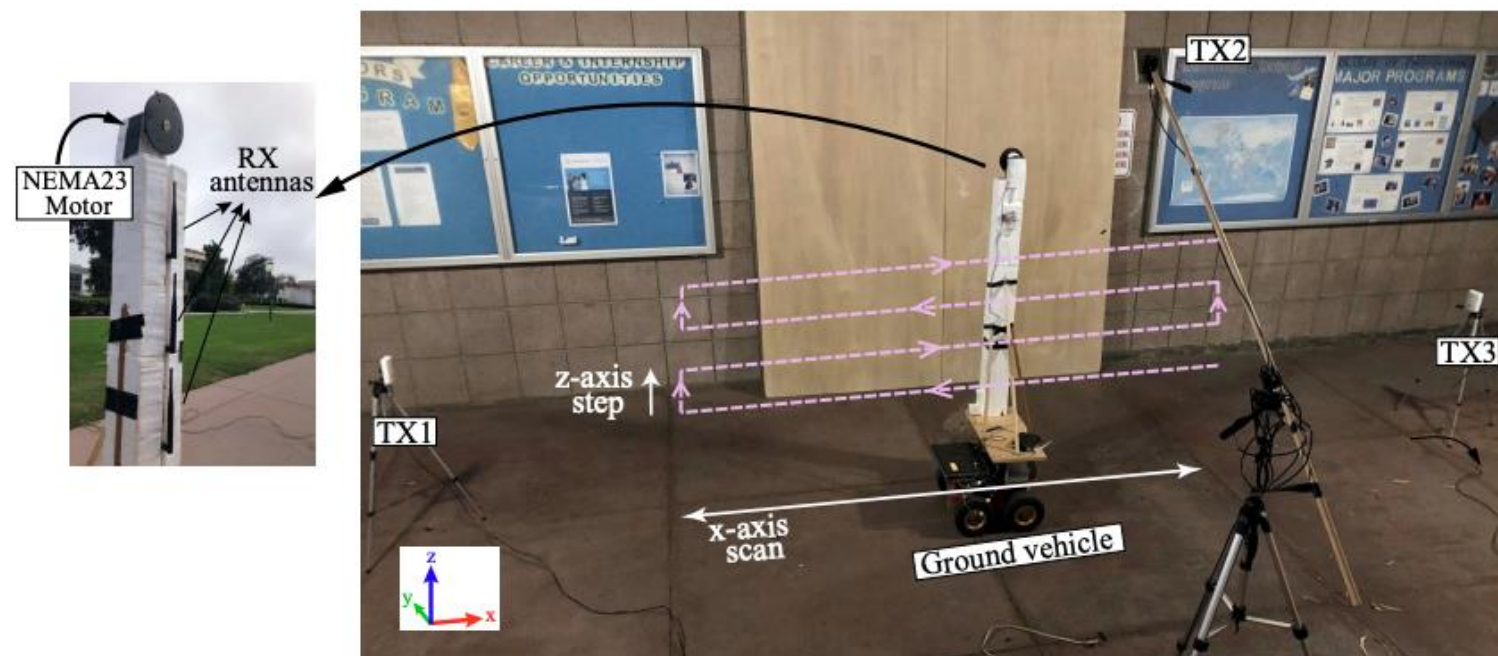
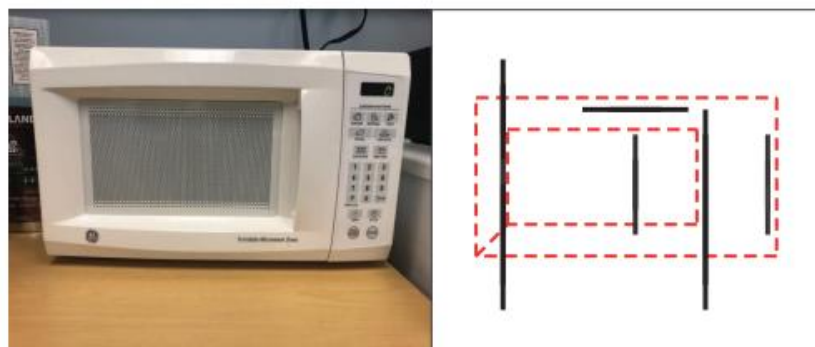


Figure 5: Sample experimental setup: 6 antennas of two laptops serve as receivers while a WiFi card of one laptop is used for transmission. A vertical structure carrying the RX antennas is mounted on a ground robot to synthesize an RX grid in the x-z plane, on which we measure WiFi CSI power measurements from three TX antennas (of one WiFi card) simultaneously.

WiFi Imaging and Pose

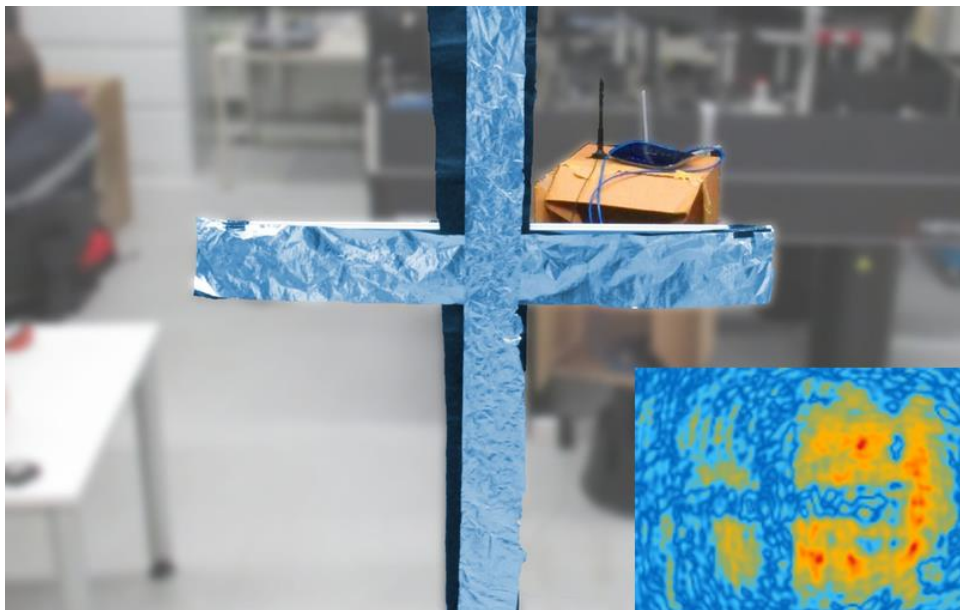
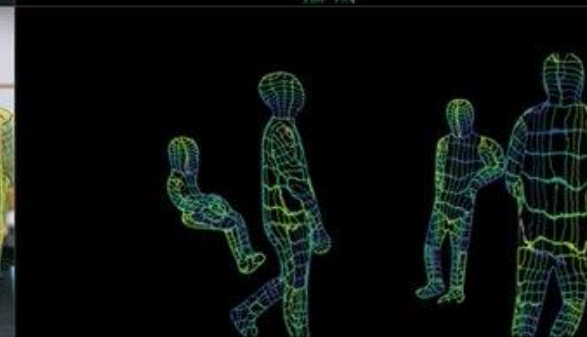
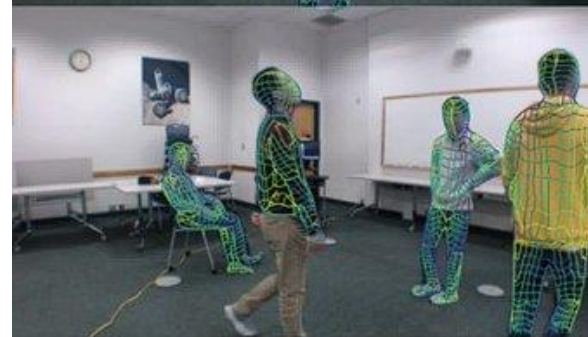
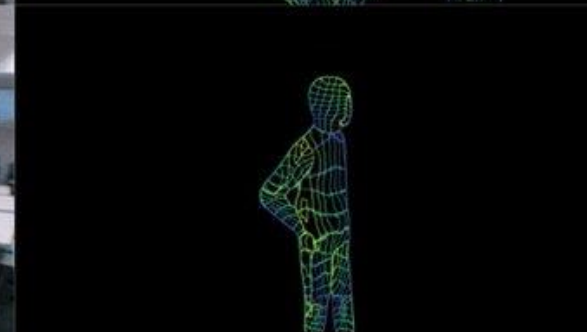
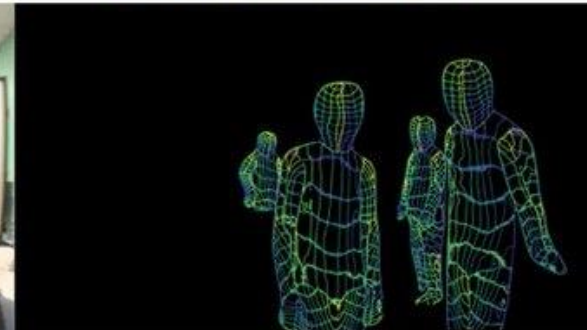


image based DensePose

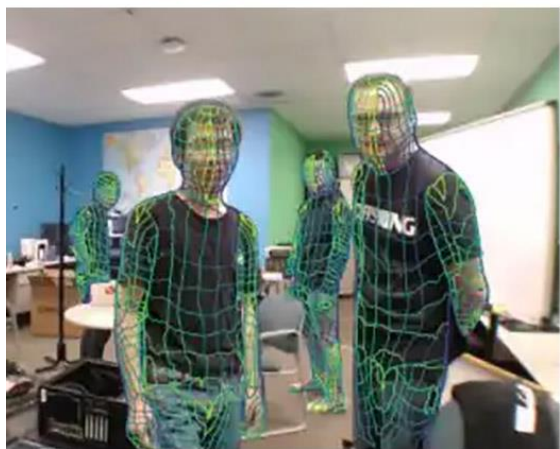


WiFi based DensePose

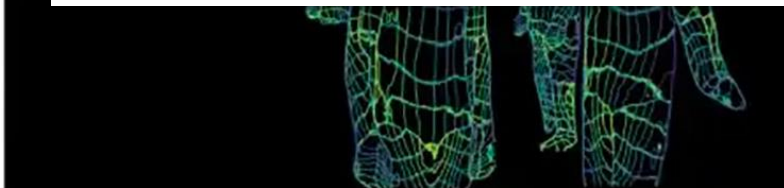


Holography of Wi-fi Radiation, Physical Review Letters, 2017
Wifract: A New Foundation for RF Imaging via Edge Tracing, MobiCom'22
DensePose From WiFi, arXiv:2301.00250v1

WiFi does more, but not everything



“People might be a little freaked out now, in the sense that internet service providers might locate what people are doing at home but, no, we are still not there. The only thing that this paper shows is that, in a very constrained setting... [with] three receivers of Wi-Fi signal, there is enough signal there for the fine-grained detection of human body parts,” said Fernando De la Torre, Carnegie Mellon University Researcher.



Sensing in 5G/6G

- Massive MIMO is standard
 - Many antennas!!
- Large bandwidths
 - e.g., 400MHz
- Meta surface (Reconfigurable Intelligence Surface, RIS)
 - Extend coverage
 - Manipulate multipath effect
- WiFi ISAC
 - Radar mode + comm mode (hardware)
 - Radar mode + comm mode (time sharing)
 - New waveform design
 - Existing/Legacy WiFi

Wellbeing Monitoring

Home Automation

Gesture Control

Motion Detection

Gait Recognition

Proximity Detection

Sleep Monitoring

Passive Tracking

WiFi Only

The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.
-- Mark Weiser (1991)

Fall Detection

Multipath: Foes or Friends?



Foes

Classical
Communications



Frenemy

Prior Wireless Sensing



Friends

Wireless AI

#WiFiCanDoMore

- Reading material

- C. Wu, B. Wang, O. C. Au and K. J. R. Liu, "Wi-Fi Can Do More: Toward Ubiquitous Wireless Sensing," in *IEEE Communications Standards Magazine*, vol. 6, no. 2, pp. 42-49, June 2022, doi: 10.1109/MCOMSTD.0001.2100111.

Questions?

- Thank you!